

Introduction to Forecasting

Course: Practical Time Series Analysis

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About This Course

- Intended for practitioners who make *real-world forecasts*.
- Focus on *short to medium-term forecasting* where statistical methods are useful.
- The mathematical level will be modest. You will need a background in *basic statistics*.
- No emphasise on proofs or technical details. Intuitive explanations of the material whenever possible.
- We will gradually introduce forecasting models by increasing the complexity of parameters taken into account.
- The programming system that we will mainly use is *Python*.

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01 What is a Forecast?

02 What is a Time Series?

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05 Patterns

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01

What is a Forecast?

def What is a Forecast?



- A **forecast** is a prediction of some future event or events.

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- A **forecast** is a prediction of some future event or events.
- Forecasts can be accurate enough or misleading. Here are examples of bad forecasts:

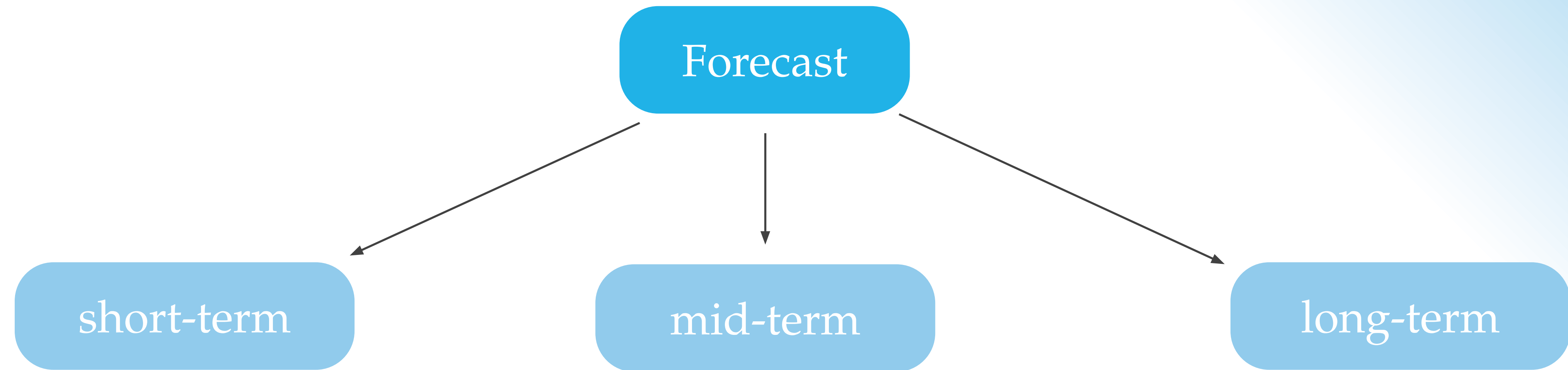
“1930 will be a splendid employment year.”

U.S. Department of Labor, New Year’s Forecast in 1929,
just before the market crash on October 29.

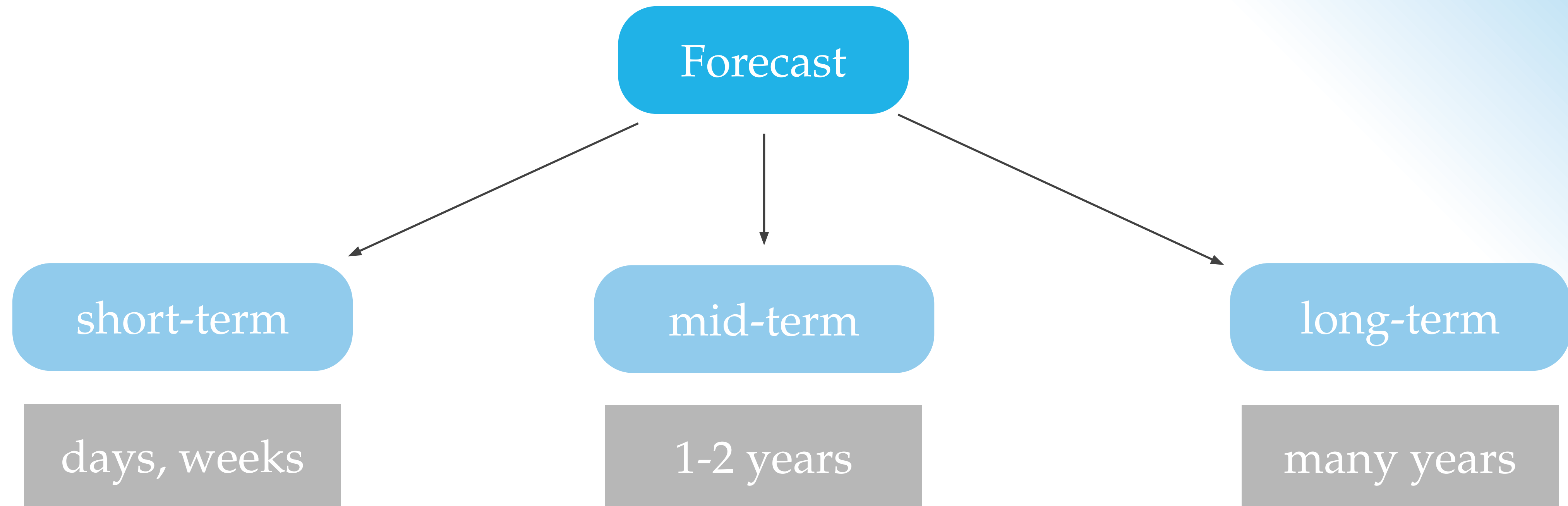
*“Computers are multiplying at a rapid rate. By the turn
of the century there will be 220,000 in the U.S.”*

Wall Street Journal, 1966.

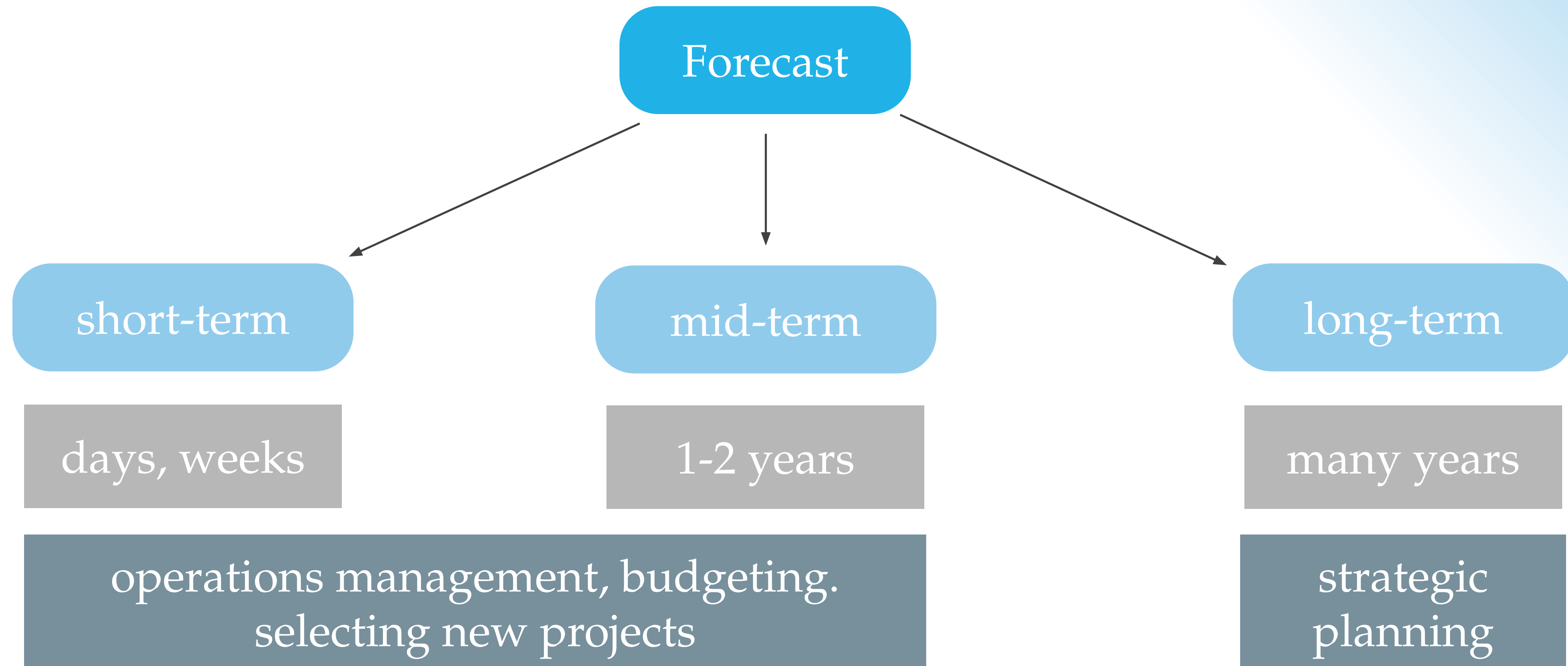
Classification of Forecasts



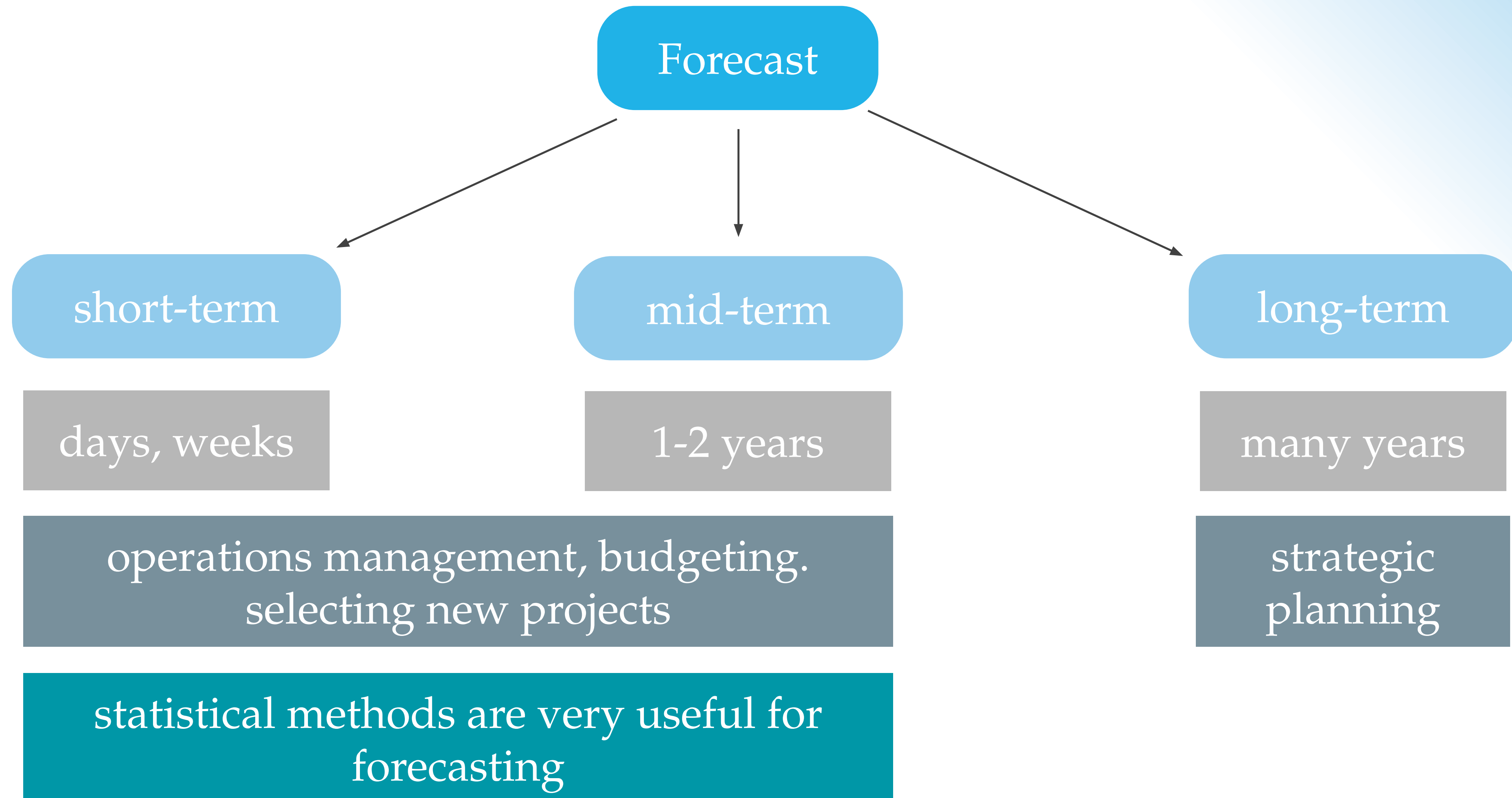
Classification of Forecasts



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Classification of Forecasts



02

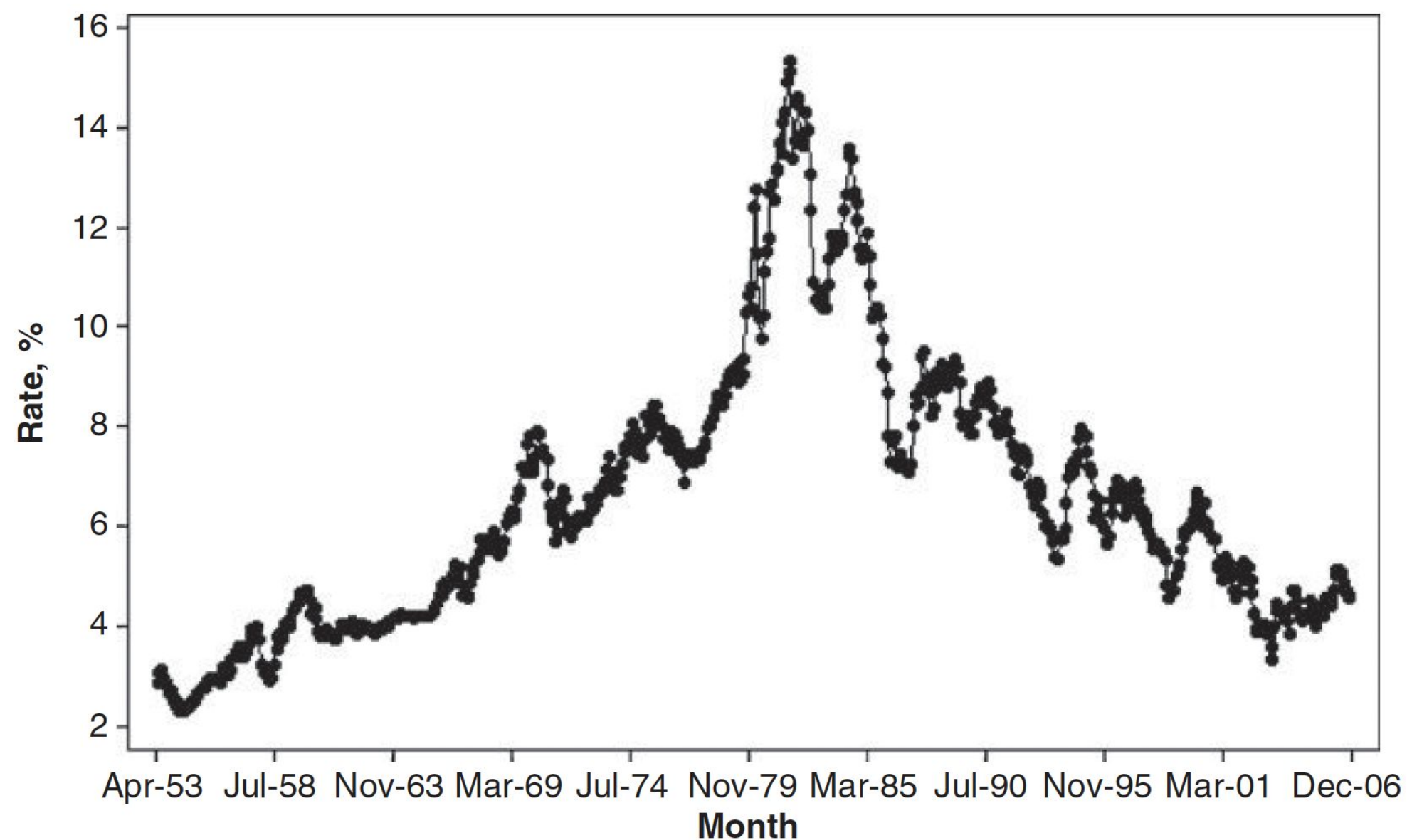
What is a Time Series?

def What is a Time Series?



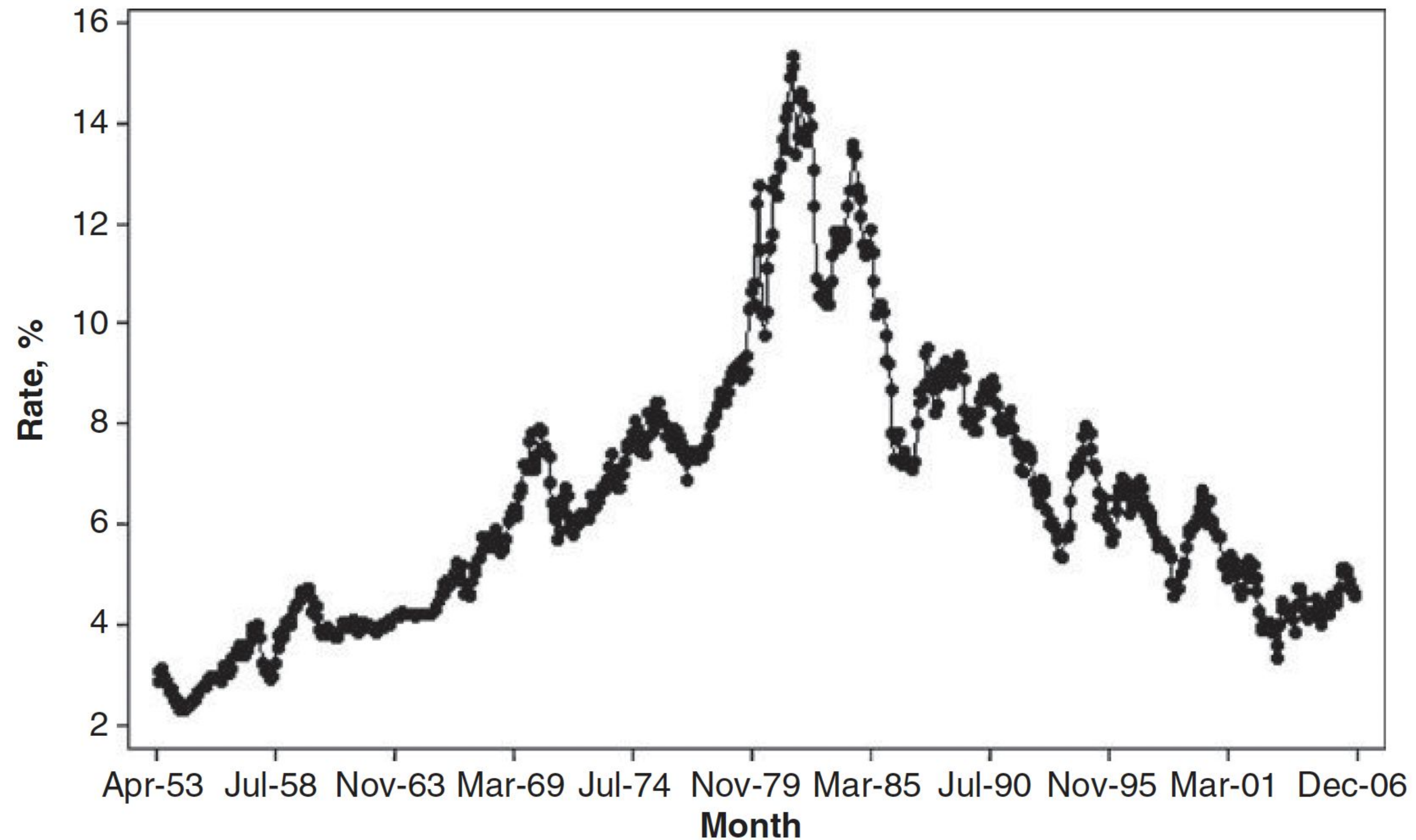
- A **time series** is a set of data points ordered in time.
- The data is *equally spaced in time*, meaning that it was recorded at every hour, minute, month, or quarter.
- Typical examples of time series include the closing value of a stock, a household's electricity consumption, or the temperature outside.

def What is a Time Series Plot?



- A **time series plot** is the visualisation of the data as a graph.
- The data may be
 - **instantaneous** (e.g. the viscosity of a chemical product at the point in time where it is measured)
 - **cumulative** (e.g. the total sales of a product during the month)
 - **statistic** that in some way reflects the activity of the variable during the time period (e.g. the daily closing price of a specific stock on the New York Stock Exchange)

Example 1: The market yield on US Treasury Securities at 10-year constant maturity



03

Applications Of Time Series

Some Applications of Time Series

Operations
Management

Marketing

Demography

Finance and Risk
Management

Economics

Biology

Astronomy

Industrial Process
Control

Meteorology

04

Important Notions

Two Types of Forecasting Techniques

Forecasting Techniques

Qualitative

- ♦ subjective in nature and require judgment on the part of experts;
- ♦ often used in situations where there is little or no historical data on which to base the forecast.

Quantitative

- ♦ make formal use of historical data and a **forecasting model**;
- ♦ the model formally summarises patterns in the data and expresses a statistical relationship between previous and current values of the variable;
- ♦ then the model is used to project the patterns in the data into the future.

Forecasting Models

Regression Models

- ♦ make use of relationships between the variable of interest and one or more related predictor variables;
- ♦ also called **causal forecasting models**.

Smoothing Models

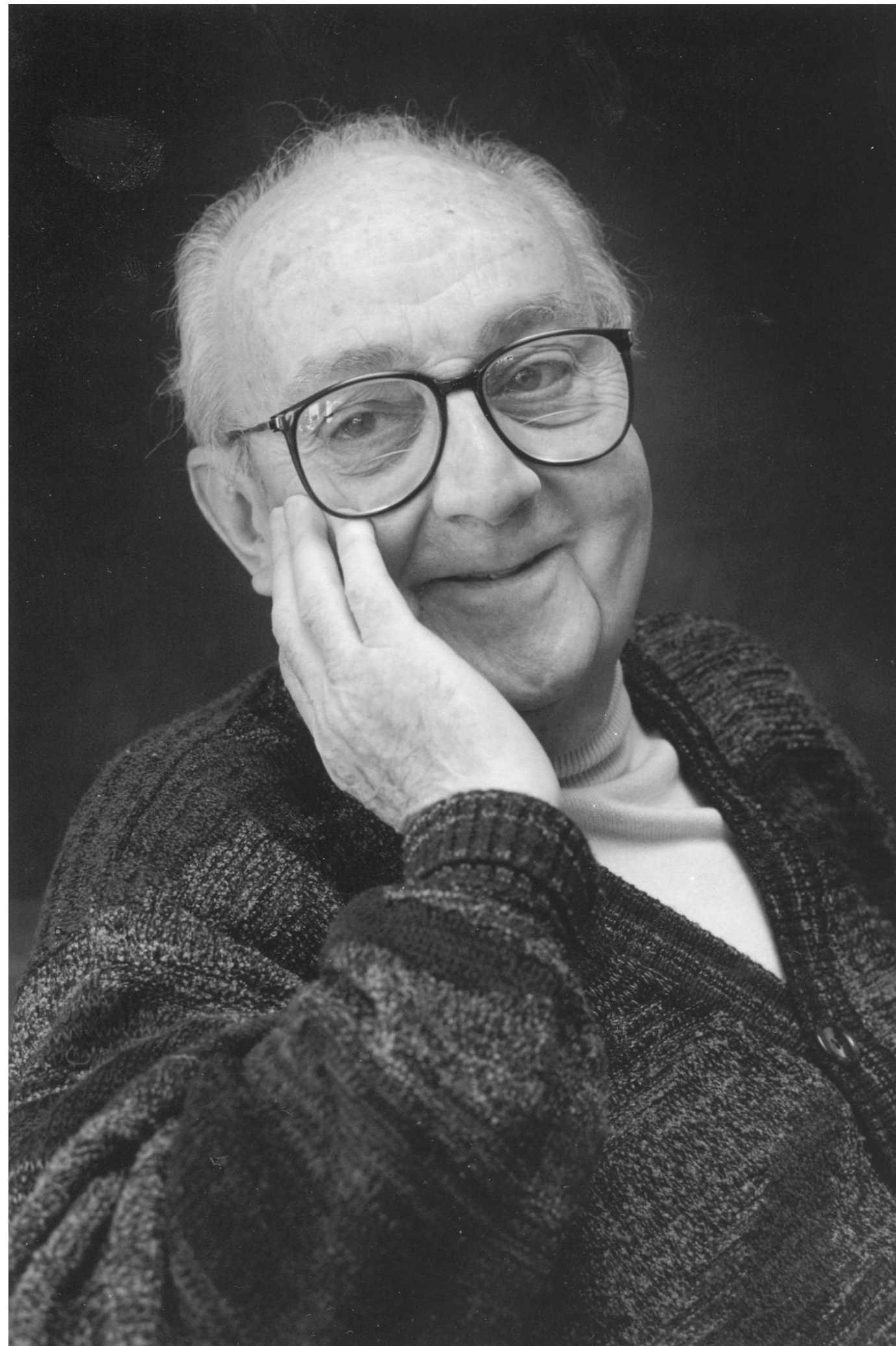
- ♦ employ a simple function of previous observations to provide a forecast of the variable of interest;
- ♦ may have a formal statistical basis, but they are often used and justified heuristically on the basis that they are easy to use and produce satisfactory results.

General Time Series Models

- ♦ employ the statistical properties of the historical data to specify a formal model;
- ♦ then estimate the unknown parameters of this model (usually) by least squares.



A bit of history...



- **George Edward Pelham Box** (18 October 1919 – 28 March 2013) was a British statistician, who worked in the areas of quality control, time-series analysis, design of experiments, and Bayesian inference. He has been called "one of the great statistical minds of the 20th century".
- He helped develop a popular time series model but was a great pragmatist. He famously said, "*All models are wrong, but some are useful*".
- Box made this statement in response to a common attitude that proper time series modelling was a matter of finding the best model to fit the data. As he explained, the idea that any model can describe the real world is very unlikely.
- One of the achievements that made George Box famous, the **Box-Jenkins method** — considered a fundamental contribution to time series analysis — appeared in 1970.

def Point Estimate, Forecast Error, Prediction Interval

- A **point estimate** (or **point forecast**) is a single number that represents our best estimate of the future value of the variable of interest.

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- A **point estimate** (or **point forecast**) is a single number that represents our best estimate of the future value of the variable of interest.
- The **forecast error** is the deviation of the point estimate from the actual value.
- The **prediction interval** is a range of values for the future observation.
- The **PI** is likely to prove far more useful in decision-making than a single number.

def Forecast Horizon, Forecast Interval, Rolling Horizon

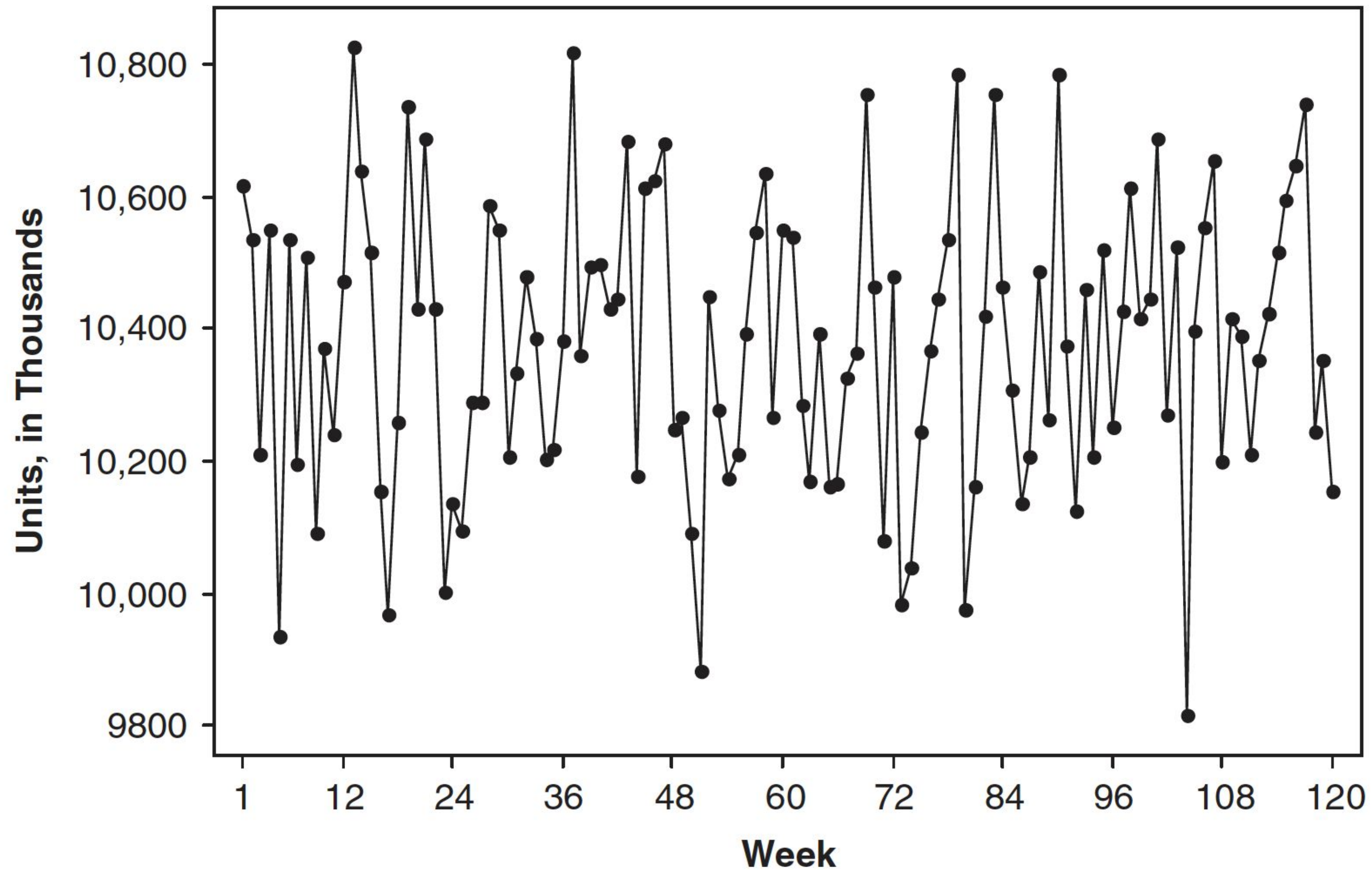
- The **forecast horizon** (or **forecast lead time**) is the number of future periods for which forecasts must be produced
- The **forecast interval** is the frequency with which new forecasts are prepared.
- If the forecast lead time is always the same length (say, T periods) and the forecast is revised each time period, then we are employing a **rolling** or **moving horizon** forecasting approach.

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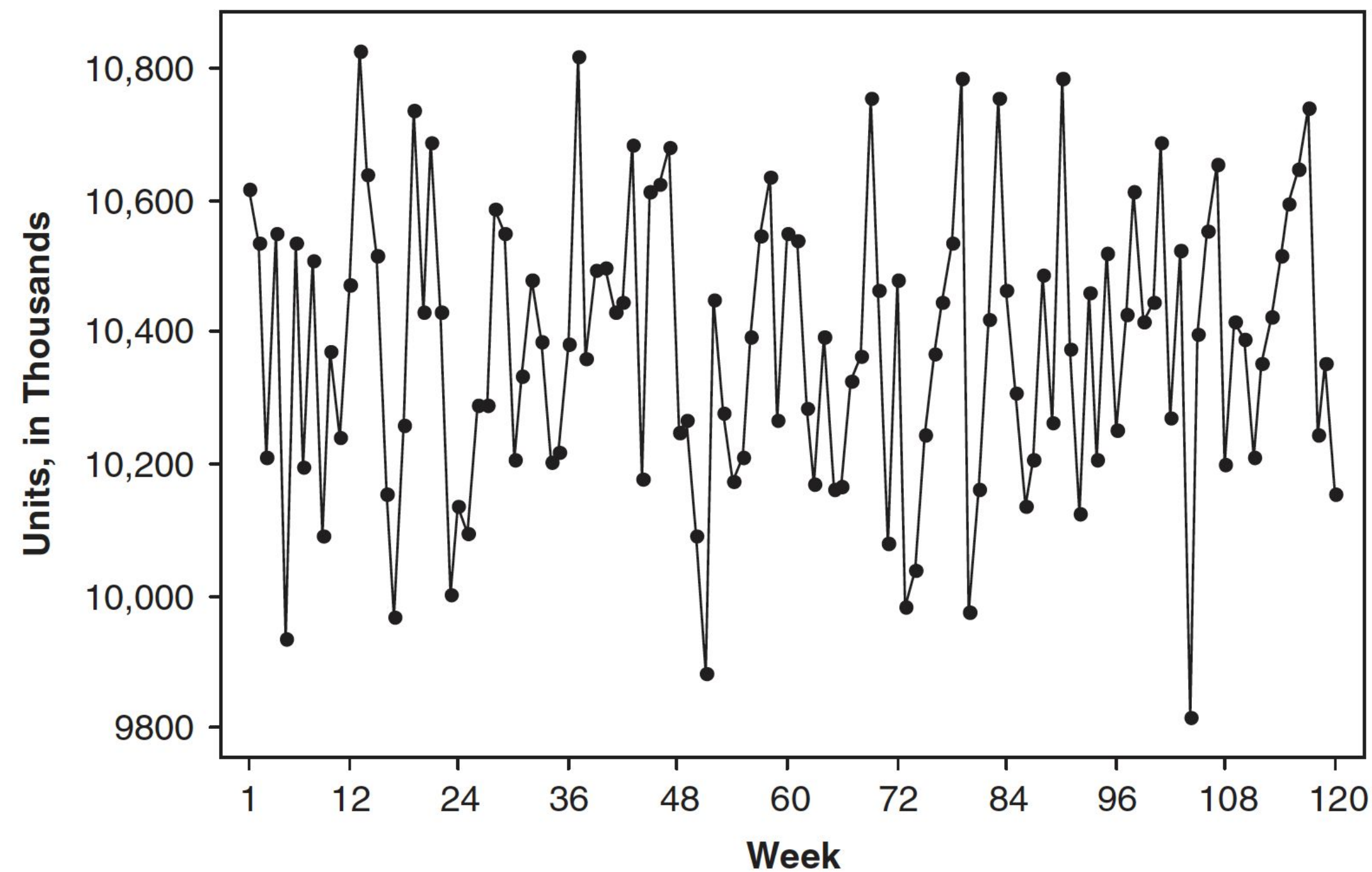
Patterns

Time series plots can reveal patterns such as *random, trends, level shifts, periods or cycles, unusual observations*, or a *combination of patterns*.

Example 2: Weekly sales of a generic pharmaceutical product

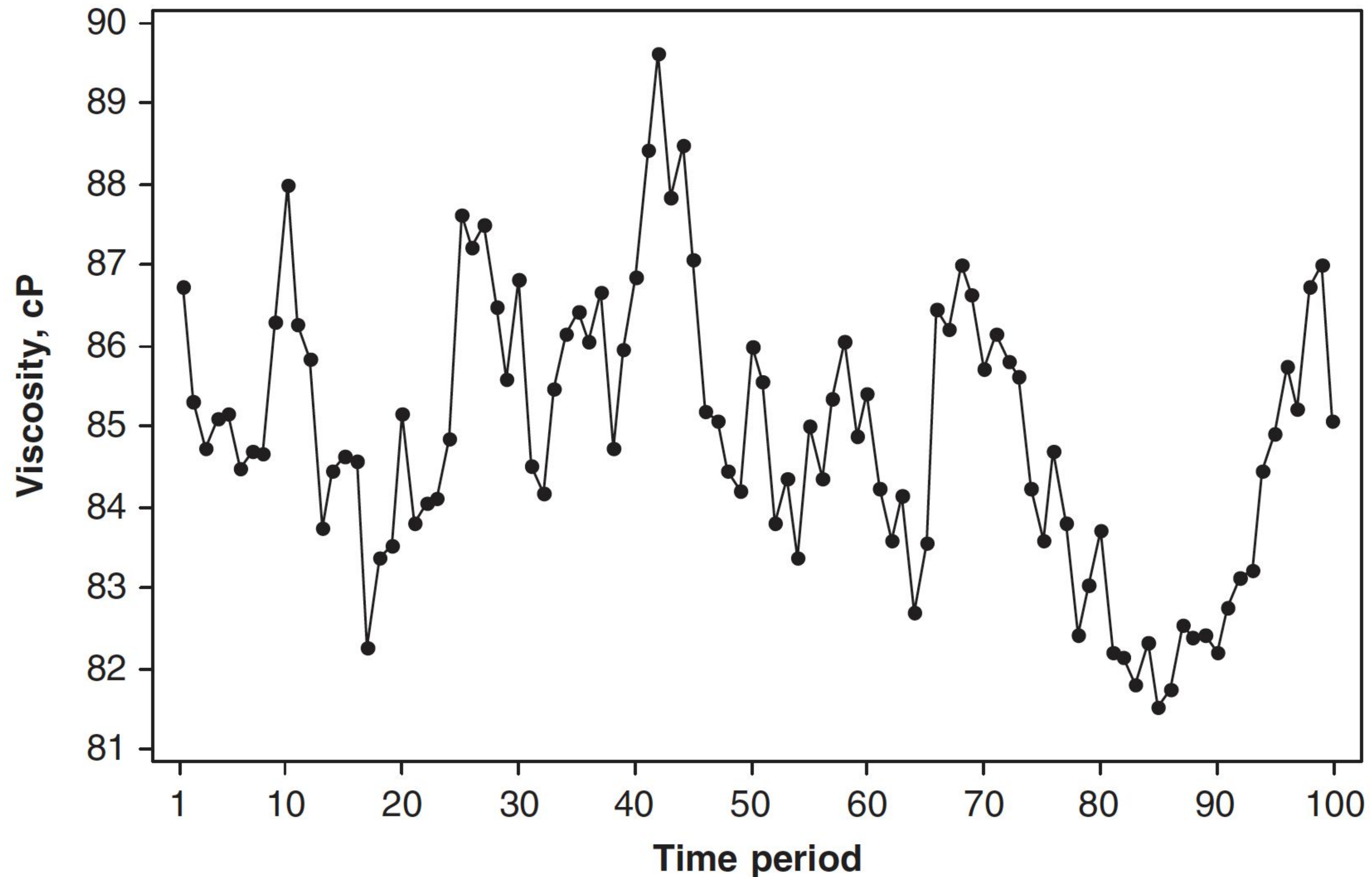


Example 2: Weekly sales of a generic pharmaceutical product

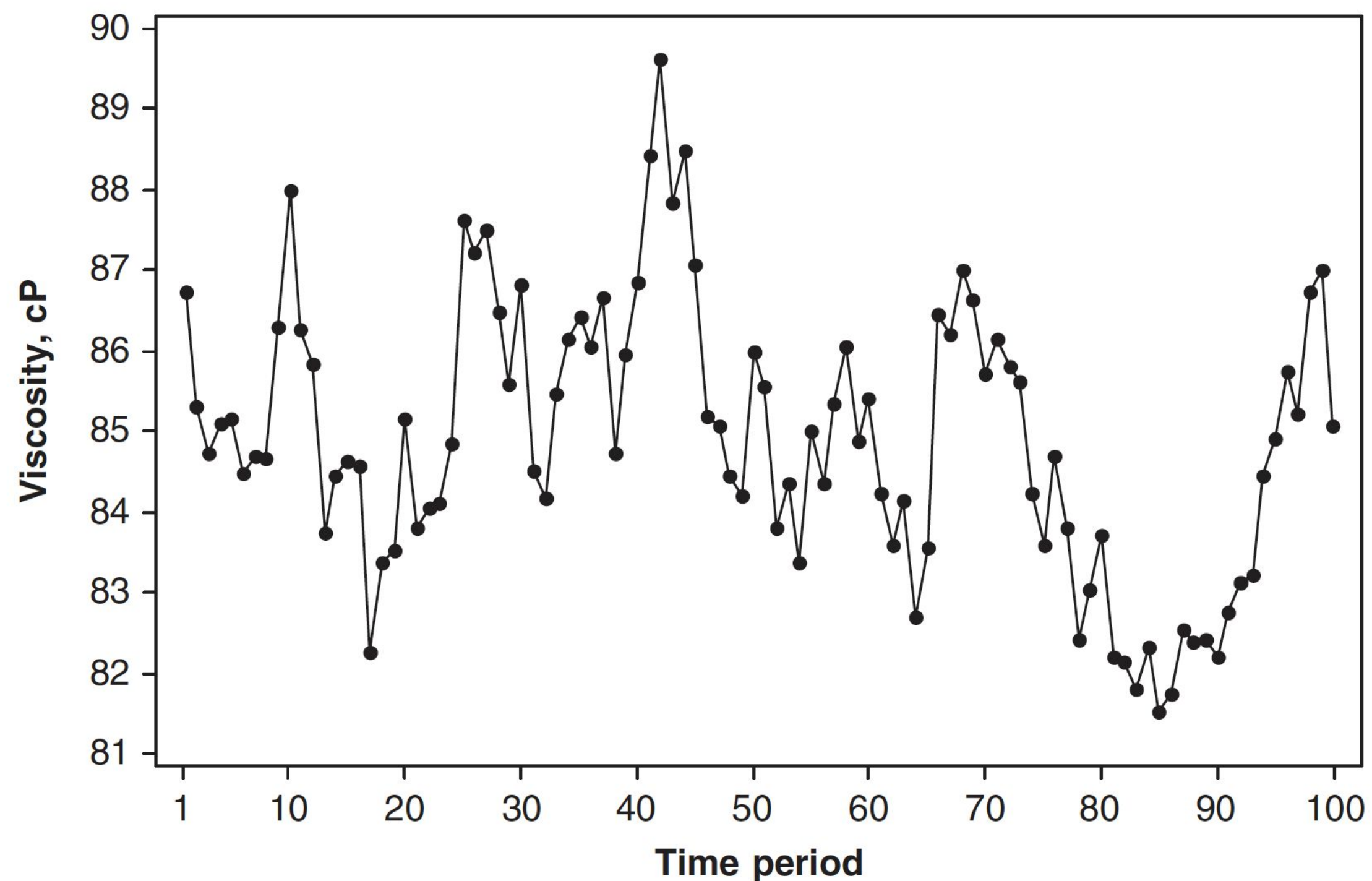


- No trend
- Constantly varying around 10,400 thousands units
- No obvious pattern
- **Random**

Example 3: Chemical process viscosity readings

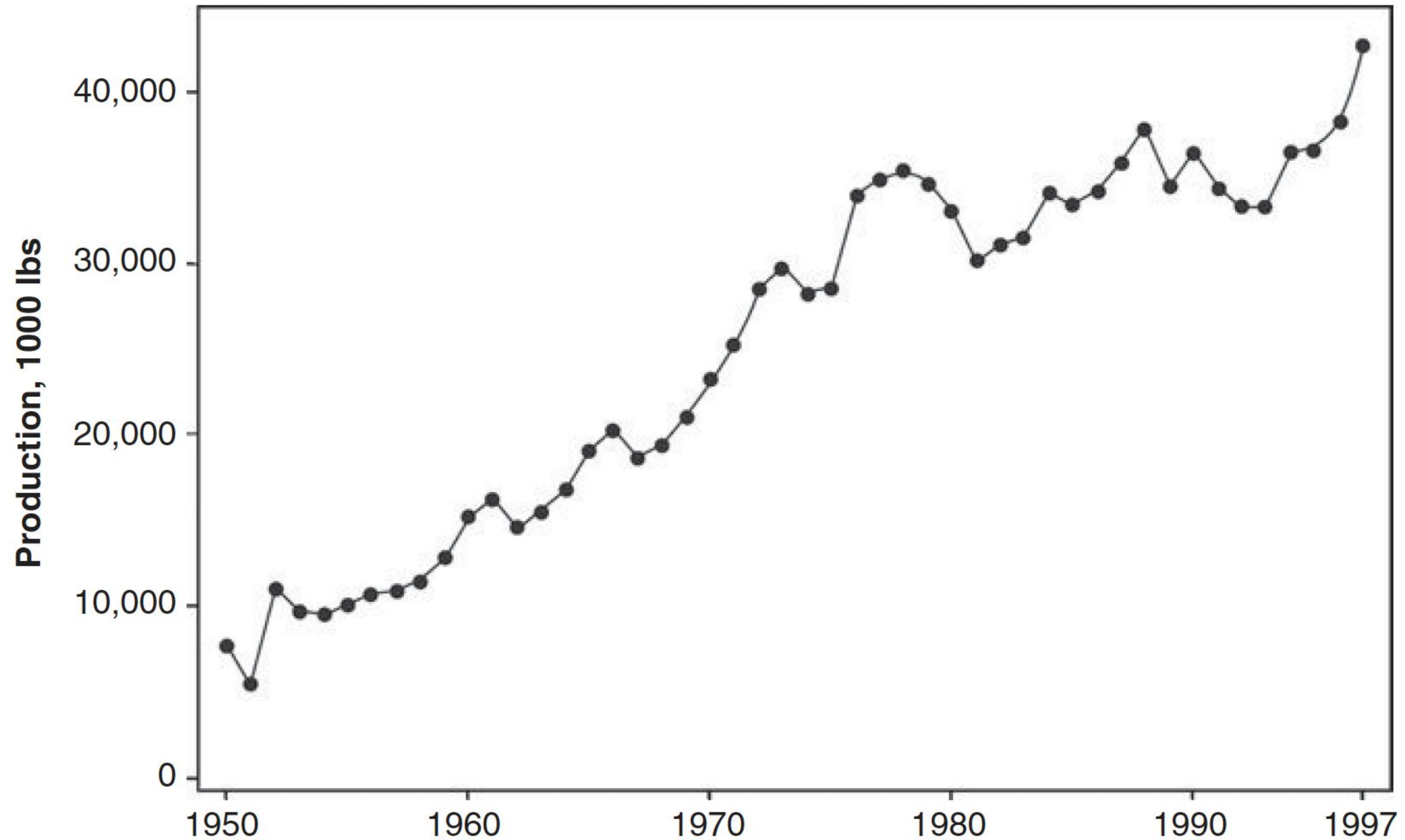


Example 3: Chemical process viscosity readings

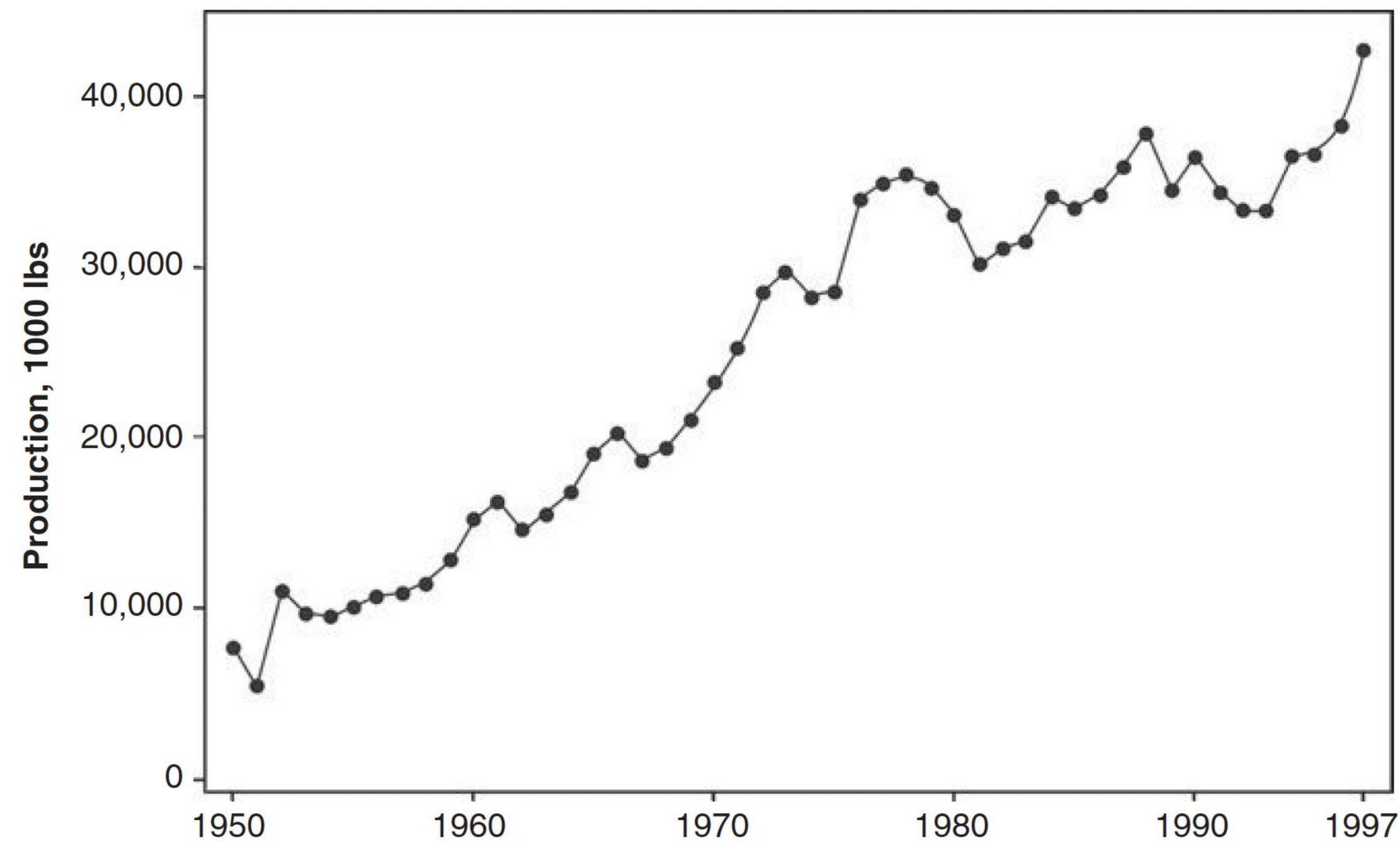


- No trend
- A long-run average of about 85 centipoises (cP)
- Some structure (not completely random)
- **Autocorrelated** — there is similarity between observations of a random variable as a function of the time lag between them
- **Positively autocorrelated** — a value above the long-run average tends to be followed by other values above the average, while a value below the average tends to be followed by other values below the average

Example 4: The US annual production of blue and gorgonzola cheeses

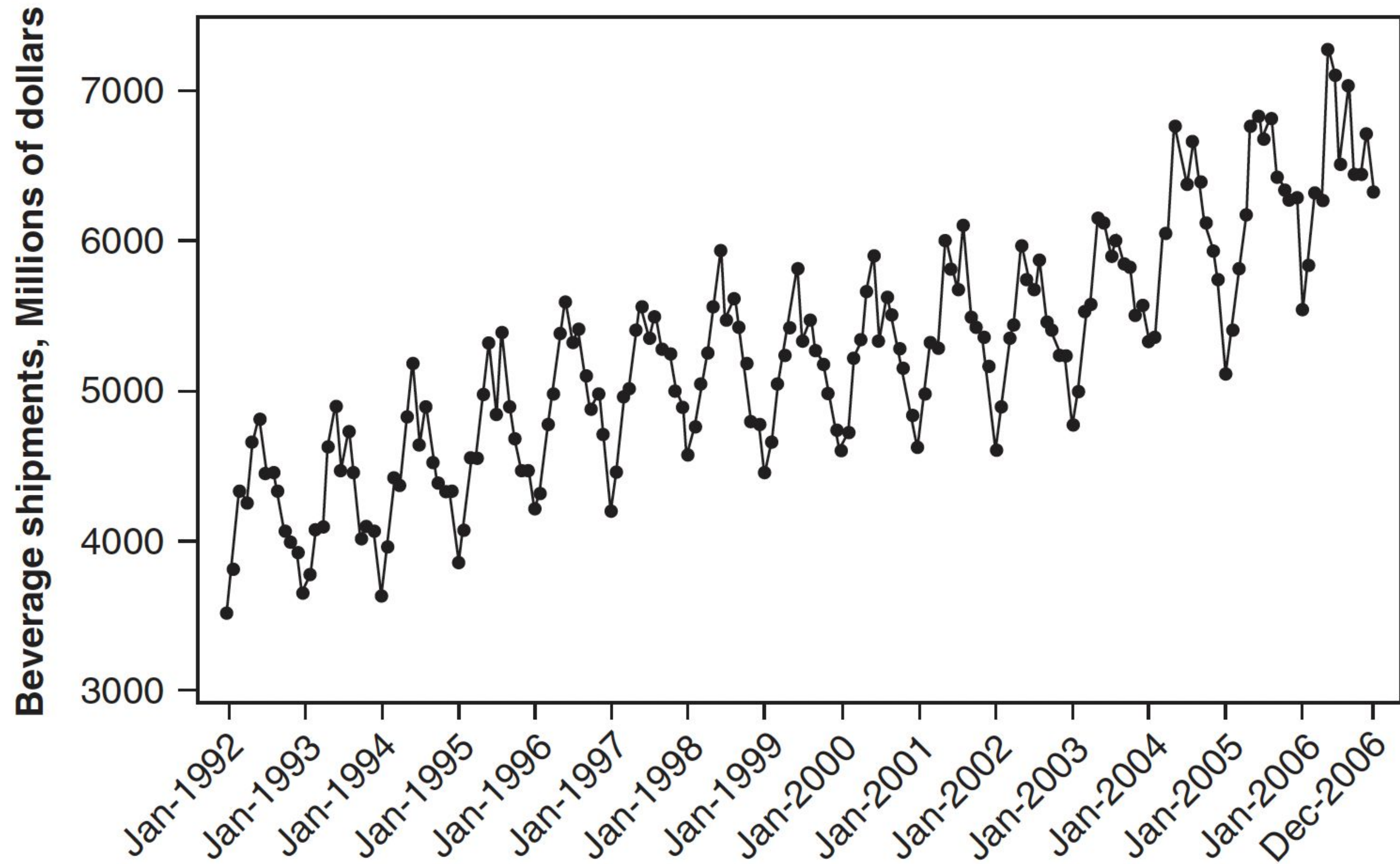


Example 4: The US annual production of blue and gorgonzola cheeses

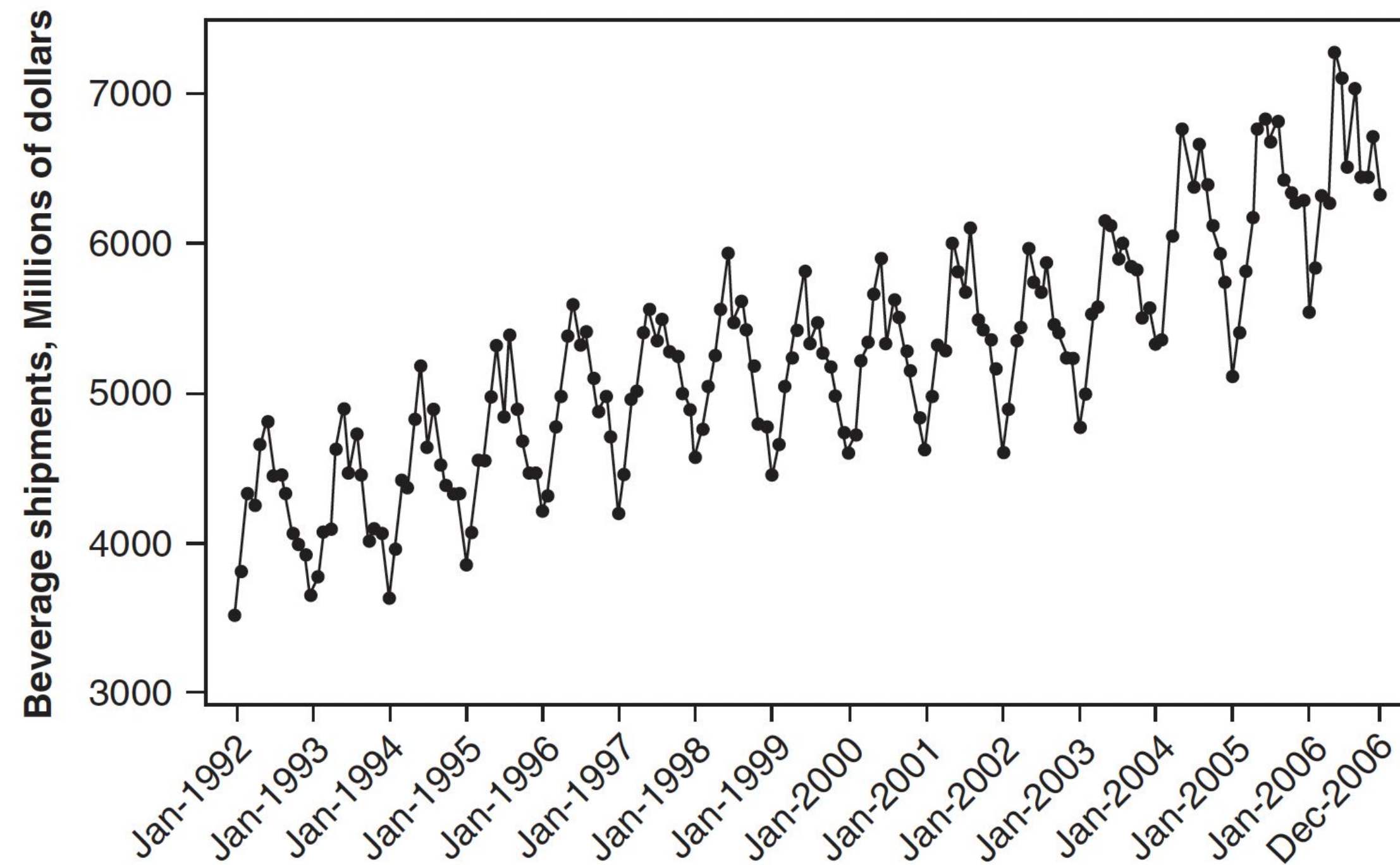


- Production quadrupled from 1950 to 1997
- **Linear trend** with a constant positive slope
- Random year-to-year variation

Example 5: The US beverage manufacturer monthly product shipments



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- Overall increasing trend
- A cyclic pattern (**seasonality**) that is repeated within each year
- Lows in January and highs in May and June

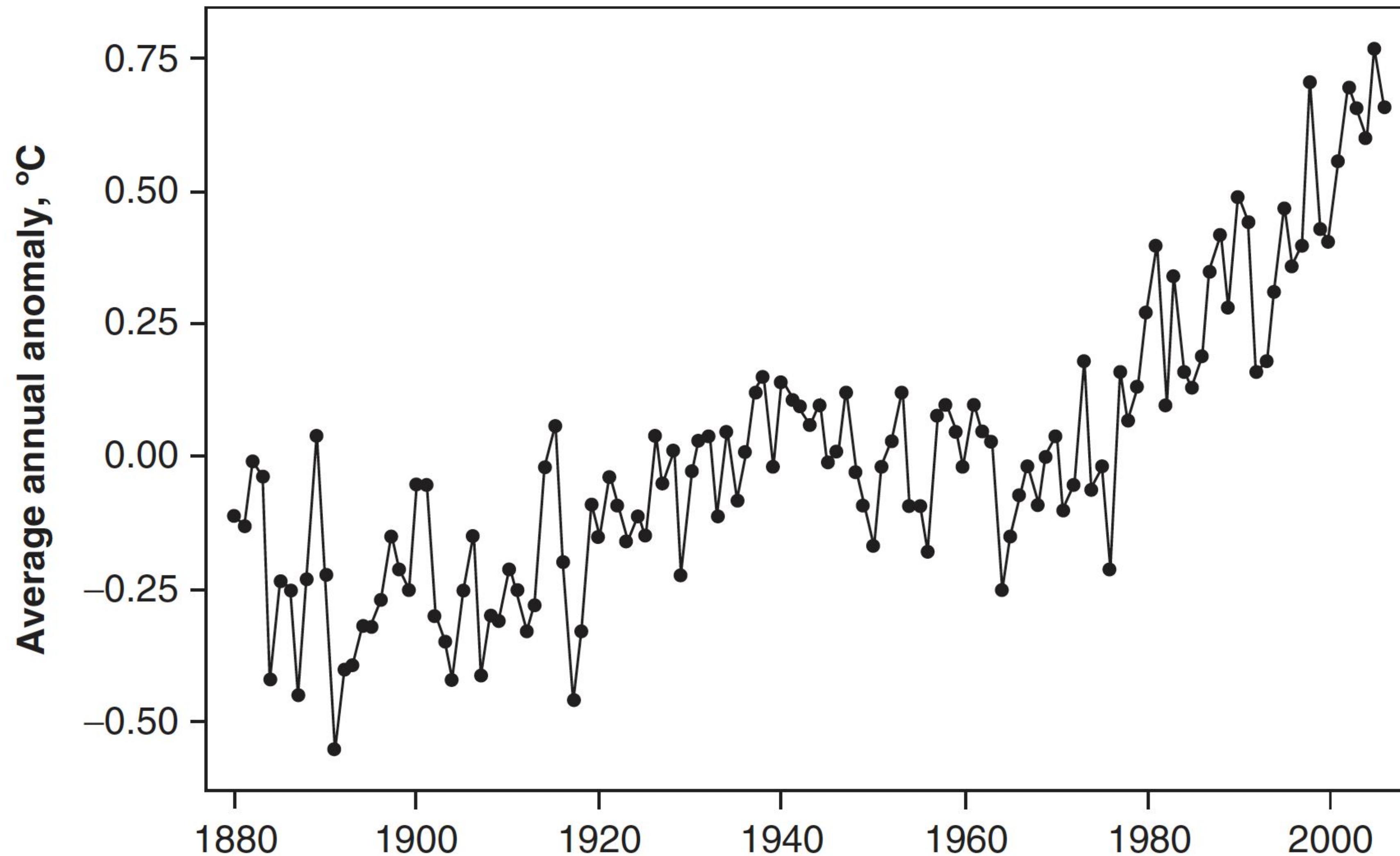


Good To Know

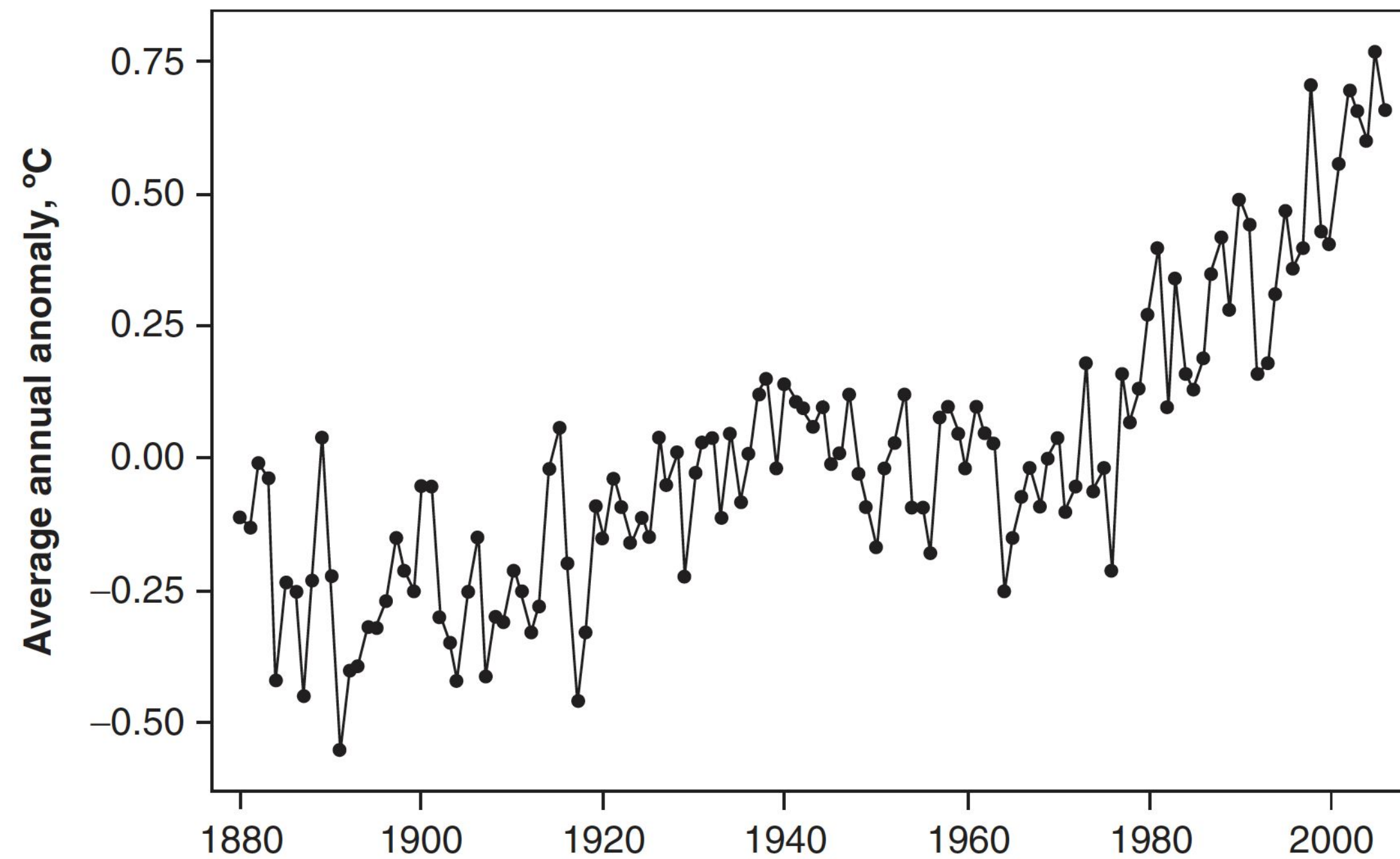


- To determine whether the Earth is *warming or cooling*, scientists look at annual mean temperatures.
- At a single station, the warmest and the coolest temperatures in a day are averaged. Averages are then calculated at stations all over the Earth, over an entire year.
- The change in global annual mean surface air temperature is calculated from a base established from 1951 to 1980, and the result is reported as an “***anomaly***”.

Example 6: Global mean surface air temperature annual anomaly

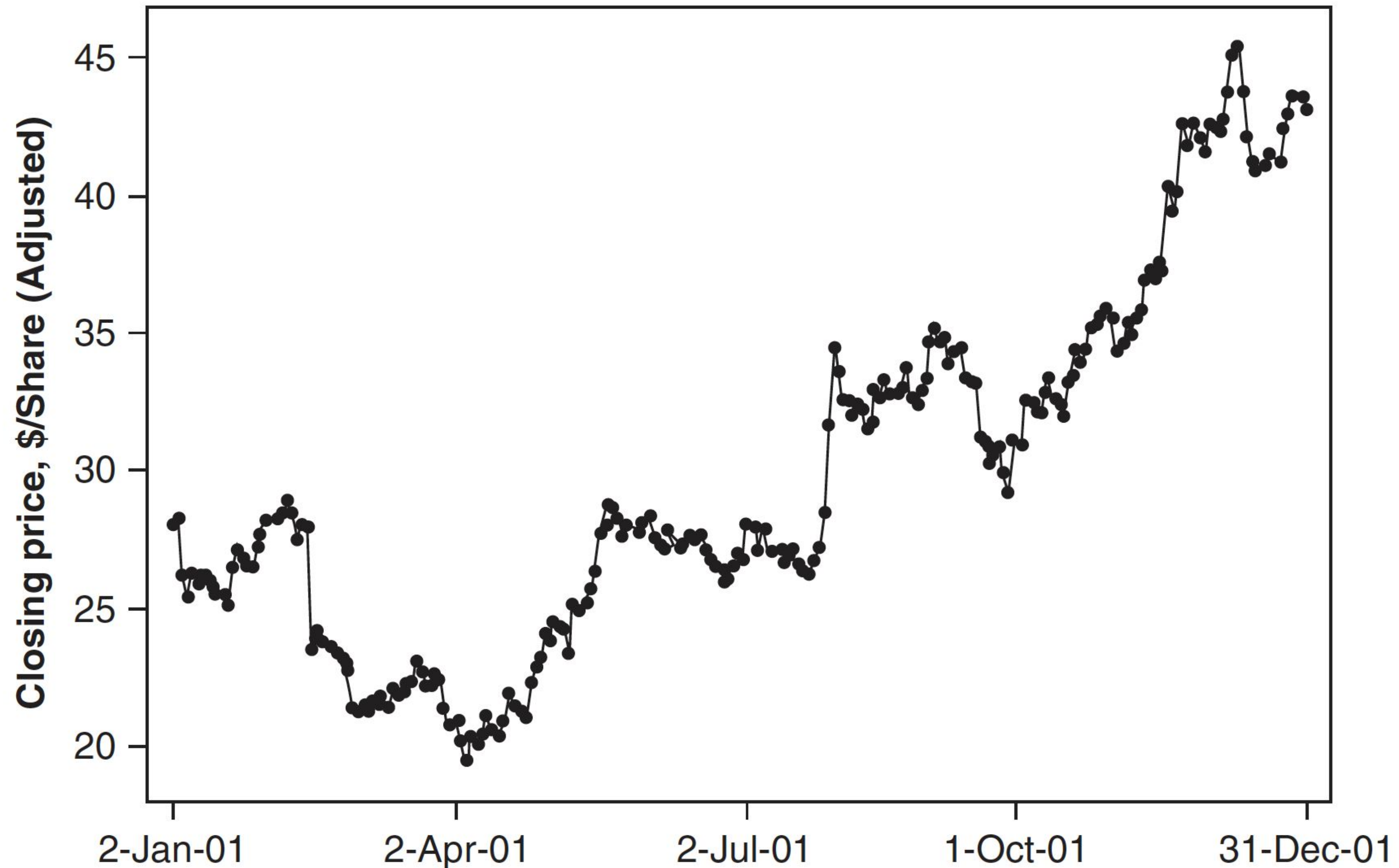


Example 6: Global mean surface air temperature annual anomaly

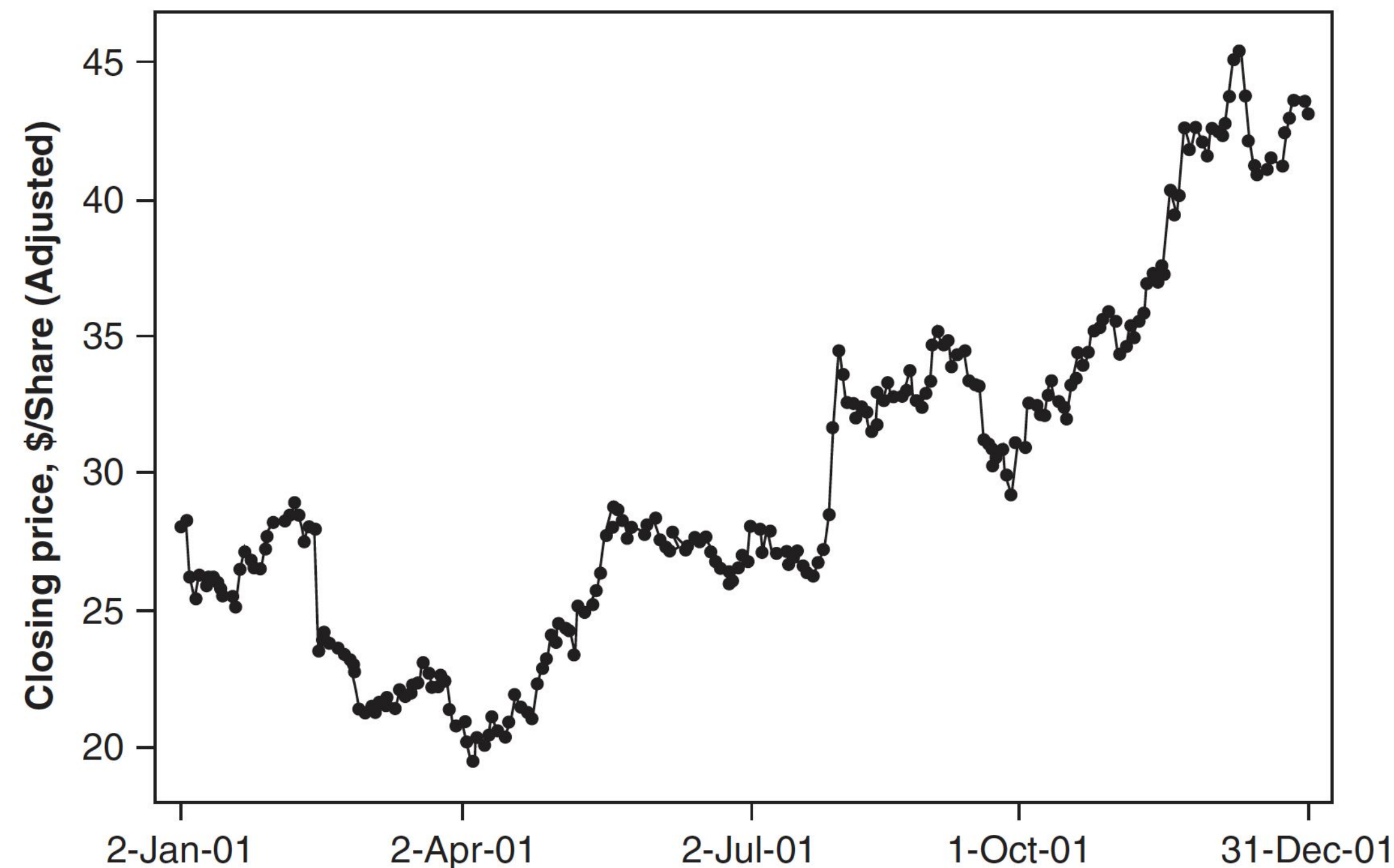


- Overall increasing trend
- Slope varies with time periods
- Slope from about 1975 to the present is much steeper

Example 7: Whole foods market stock price, daily closing adjusted for splits

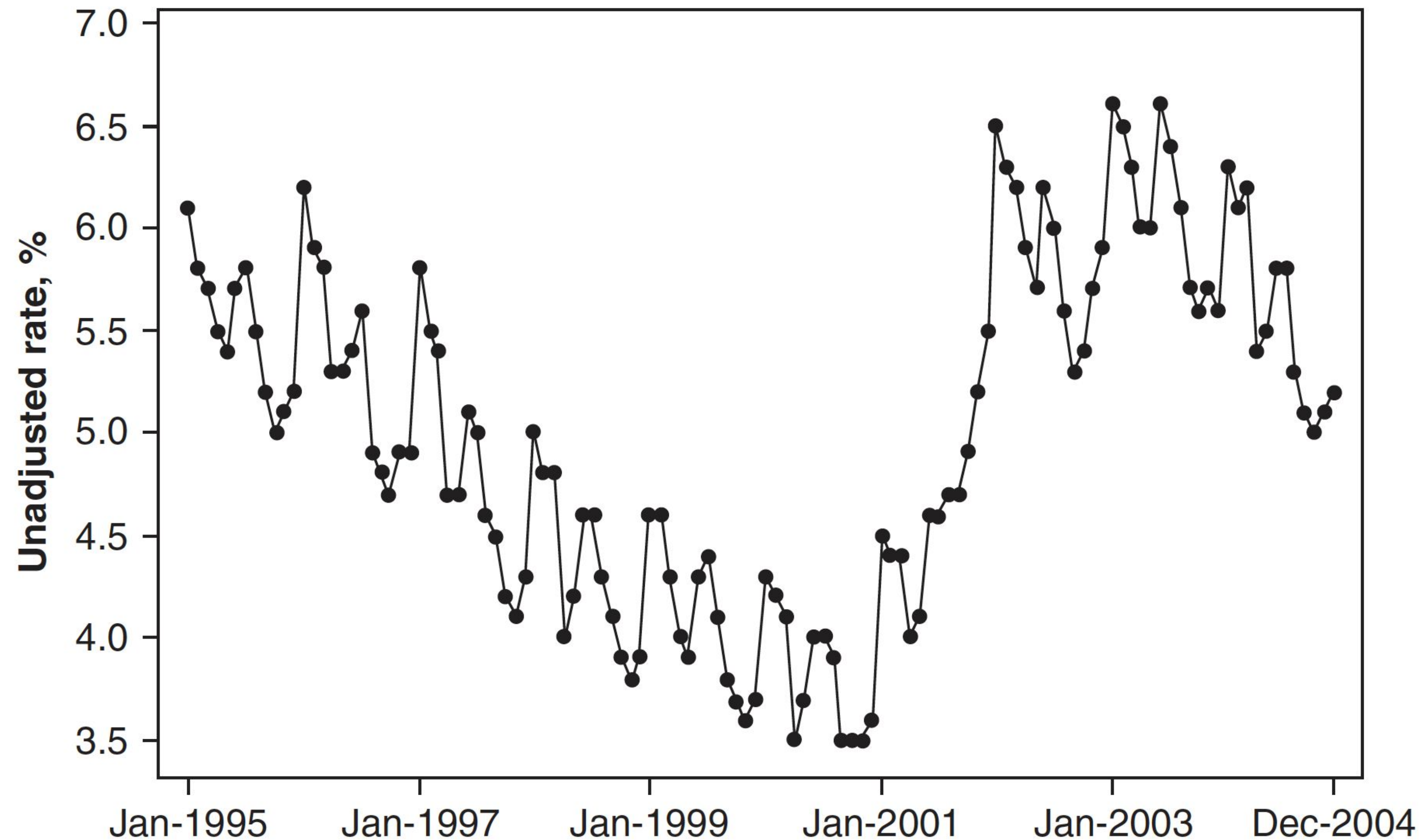


Example 7: Whole foods market stock price, daily closing adjusted for splits

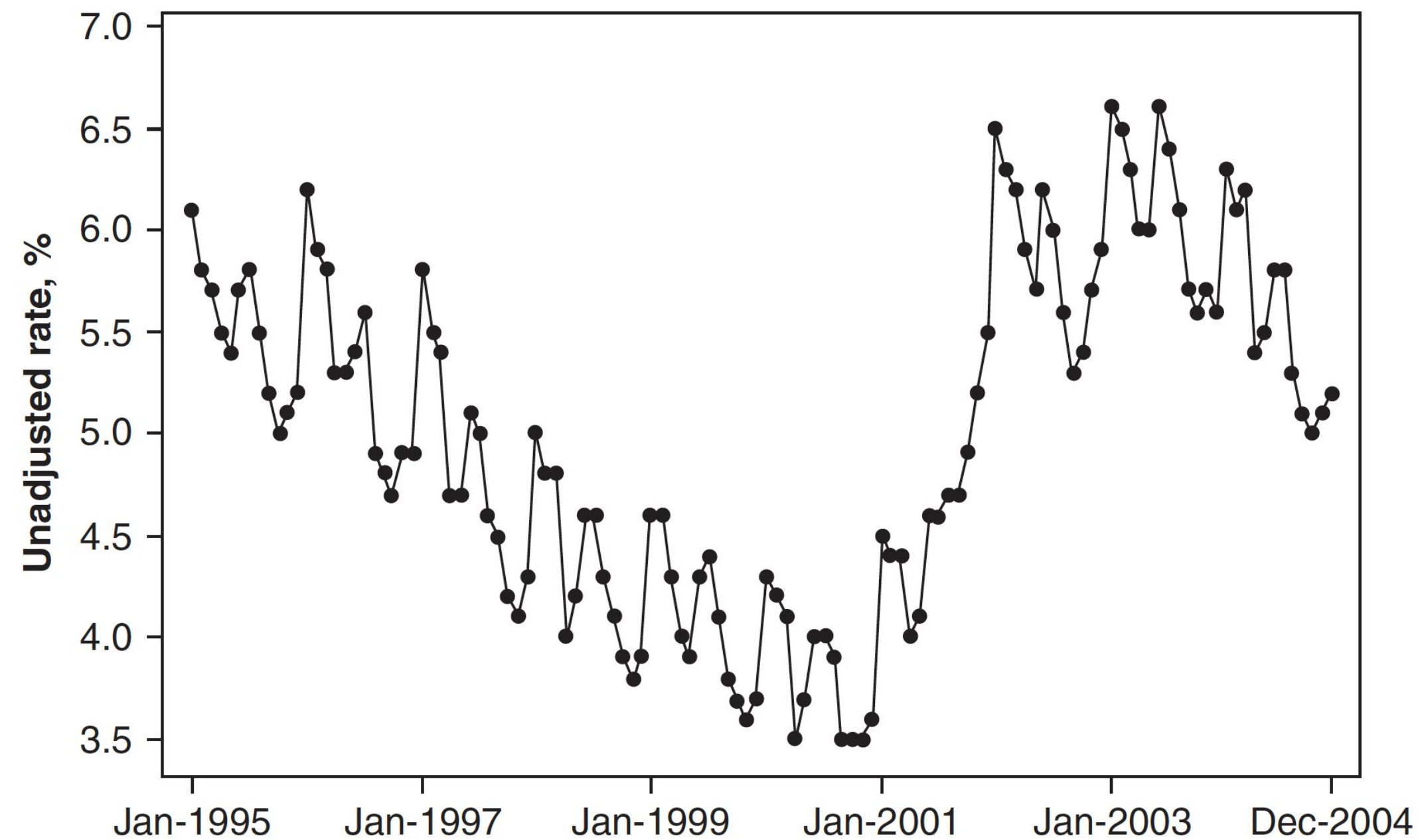


- Overall increasing trend
- Price changes at different rates, including occasional abrupt shifts in level
- **Non-stationary behaviour** — there is no consistent mean level over time

Example 8: Monthly unemployment rate (full-time labor force, unadjusted)



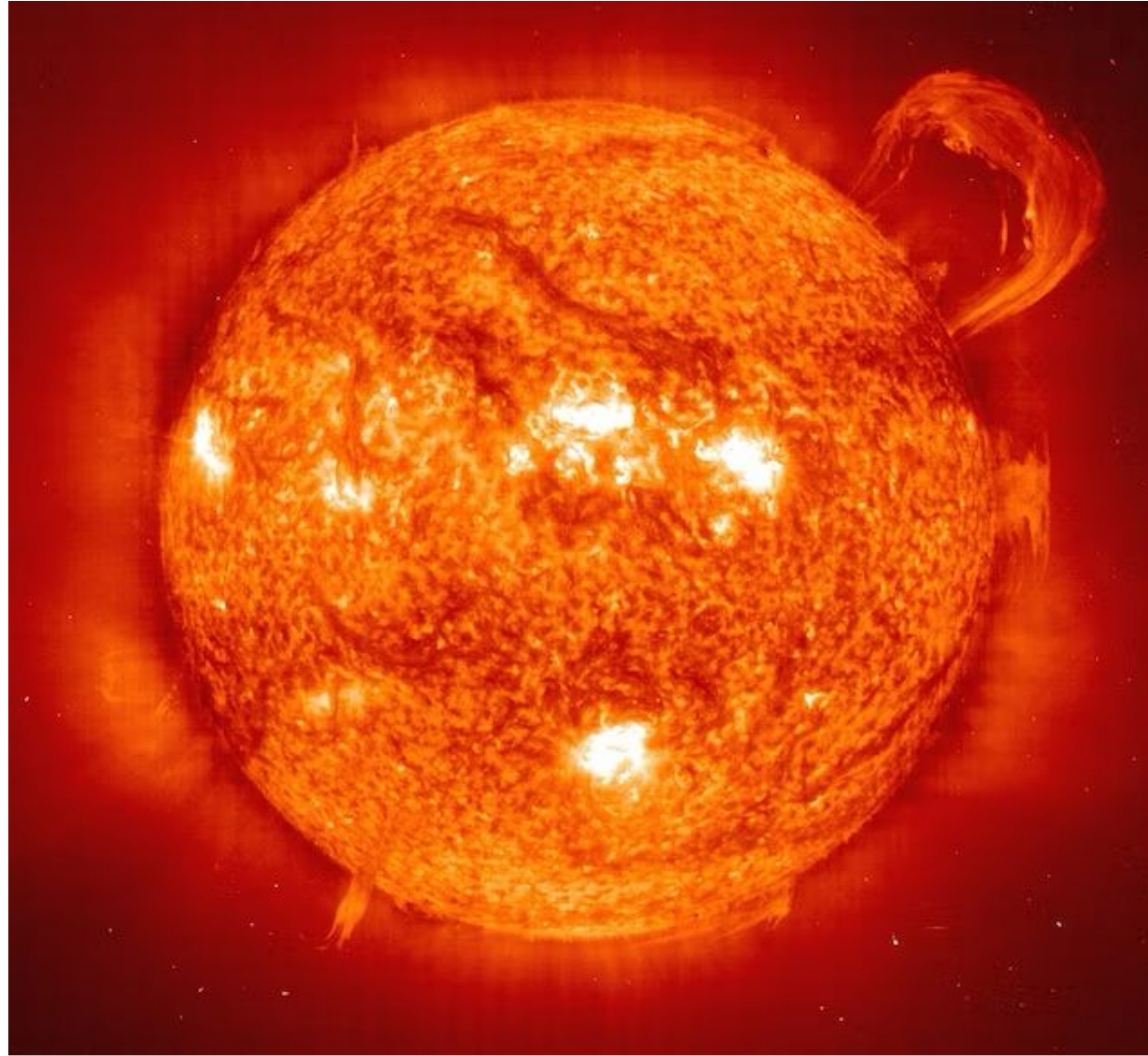
Example 8: Monthly unemployment rate (full-time labor force, unadjusted)



- A mixture of patterns
- A cyclic pattern within a year
- Highs in January, February, March
- The overall level is also changing, from a gradual decrease, to a steep increase, followed by a gradual decrease

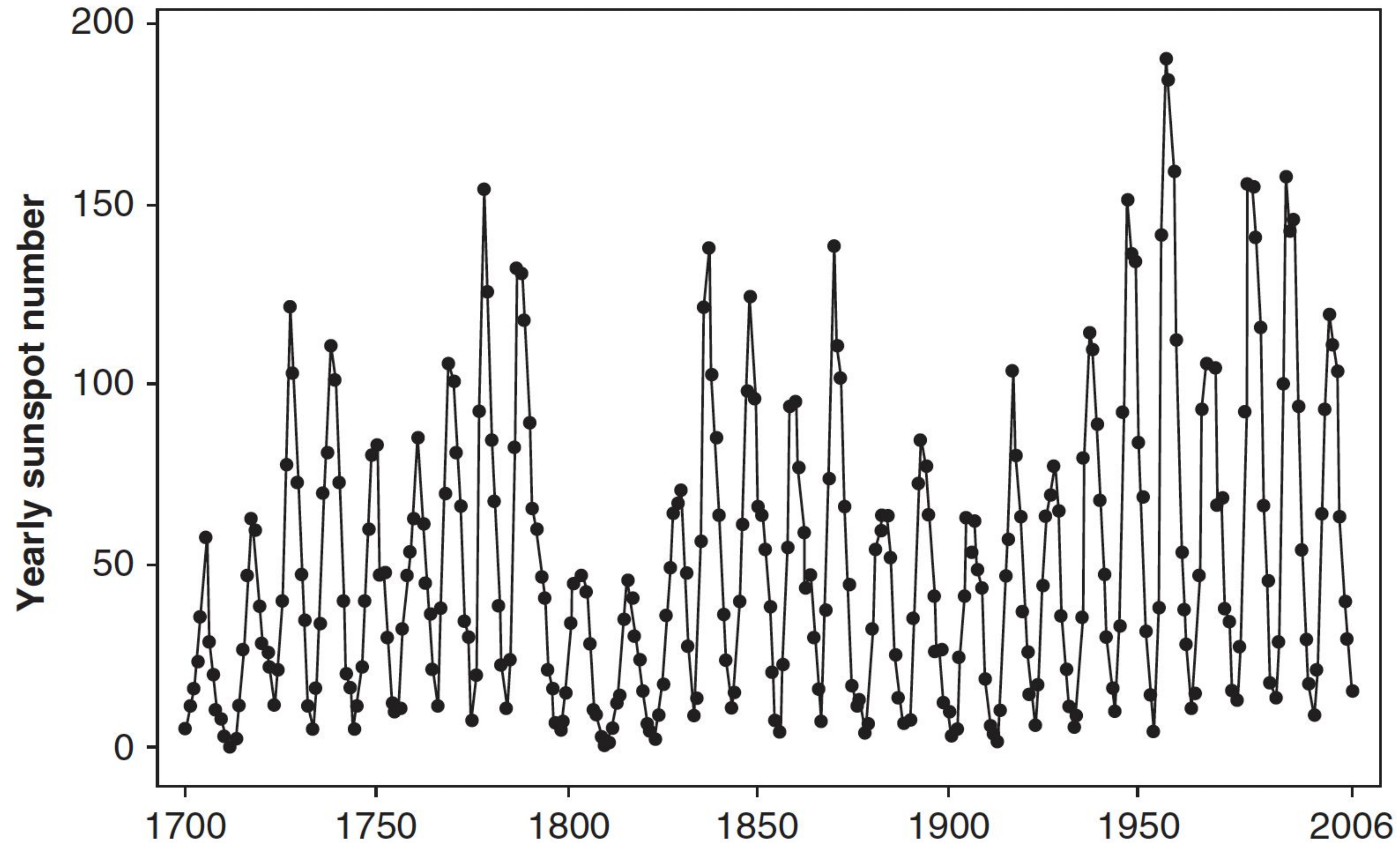


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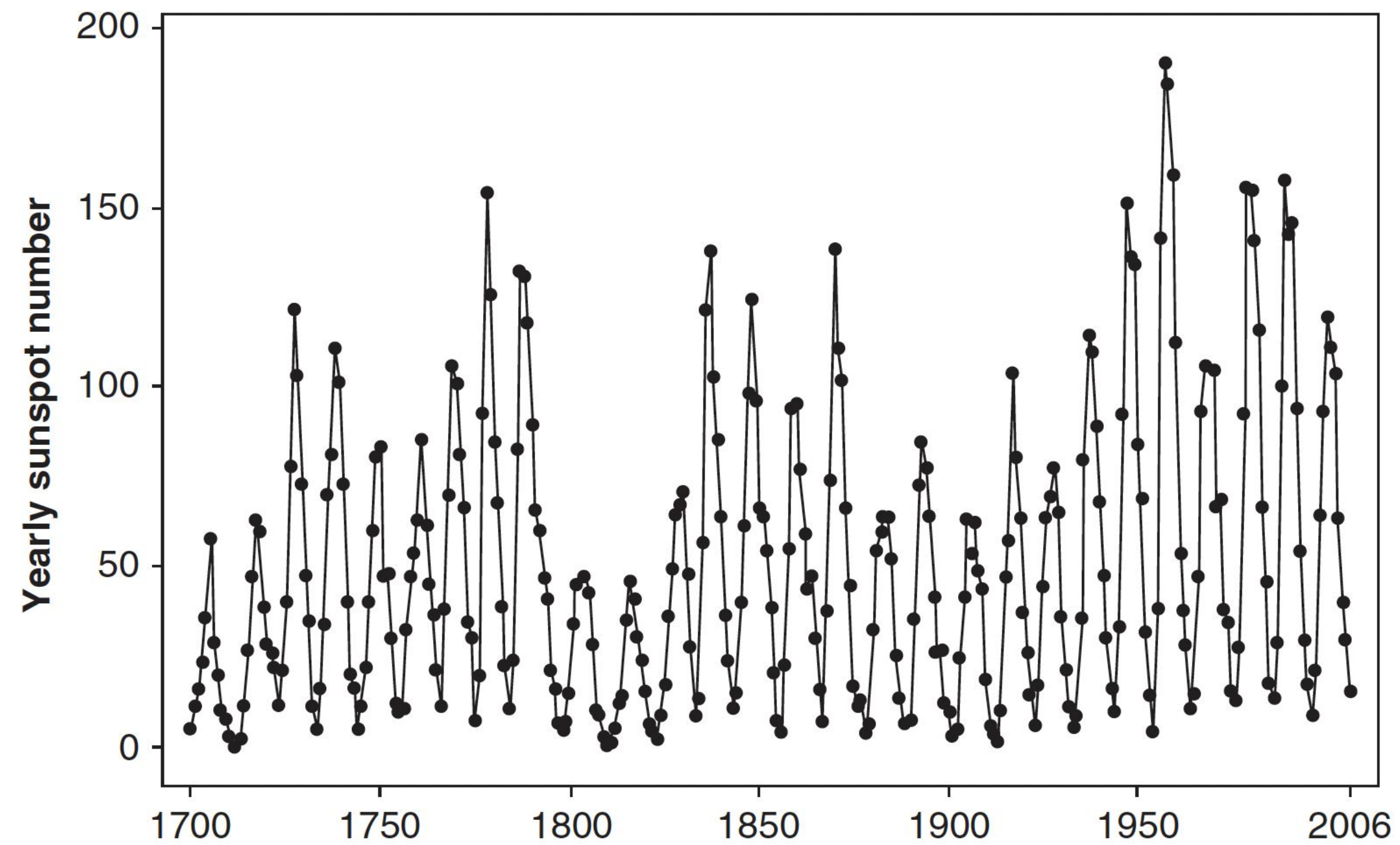


- **Solar activity** has long been recognised as a significant source of noise impacting consumer and military communications, including satellites, cell phone towers, and electric power grids.
- The ability to accurately forecast solar activity is critical to a variety of fields.
- The **International Sunspot Number R** is the oldest solar activity index.
- Sunspot time series were recorded in ancient China as early as *800 BC*, making sunspot data collection one of the most well-recorded natural phenomena ever.

Example 9: The International Sunspot Number

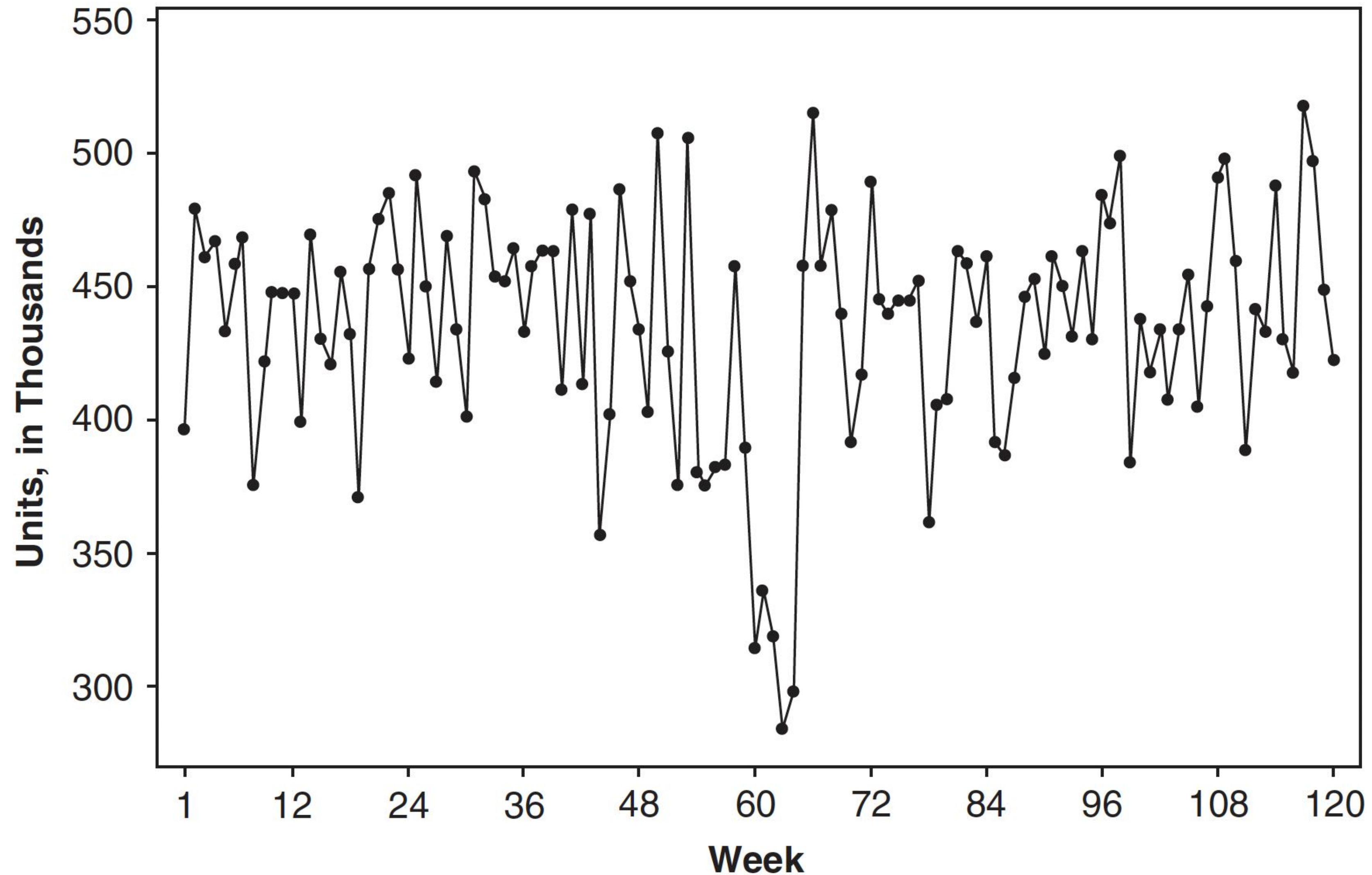


Example 9: The International Sunspot Number

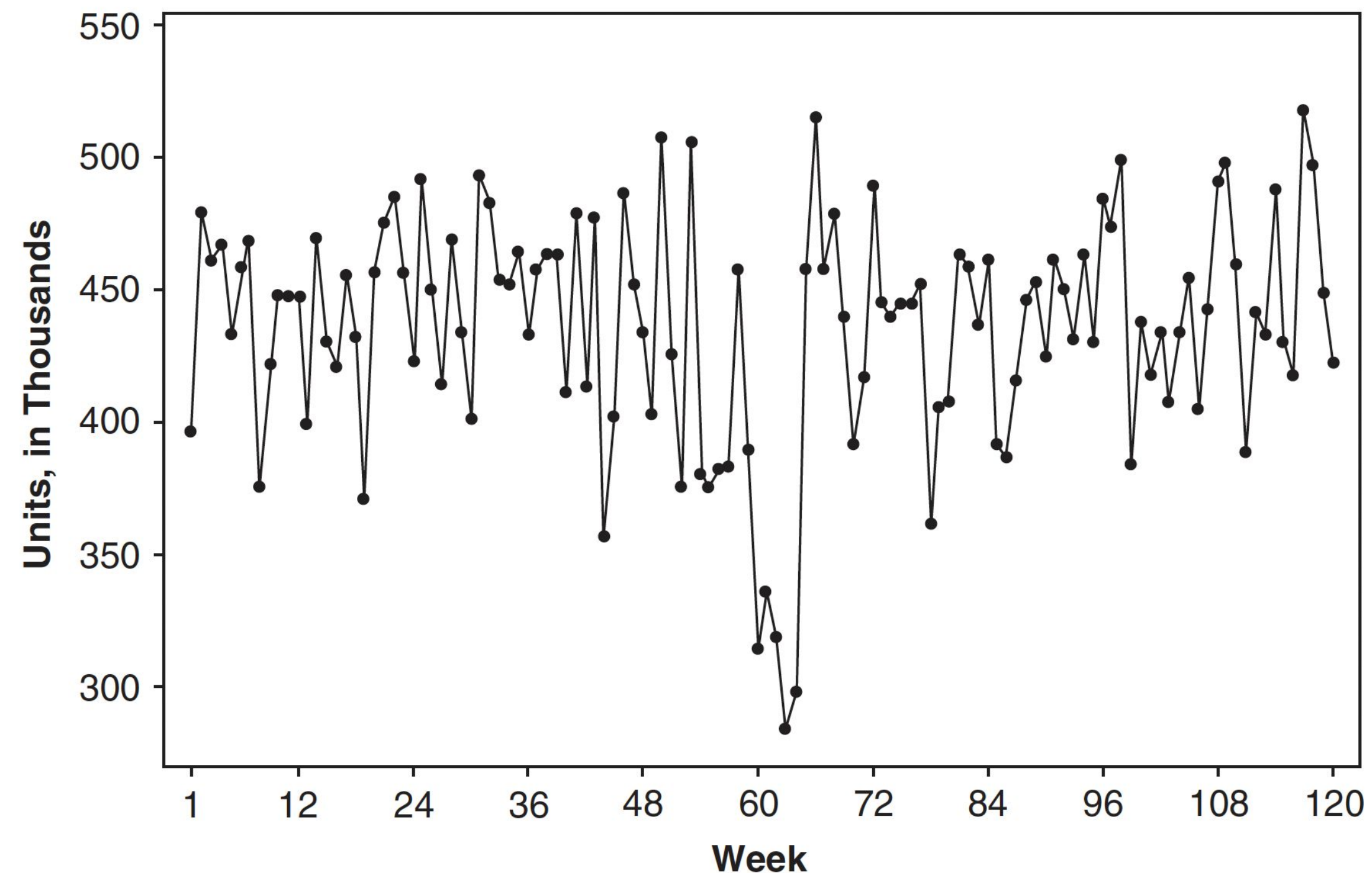


- **Cyclic patterns of varying magnitudes**

Example 10: Pharmaceutical product sales (atypical events)

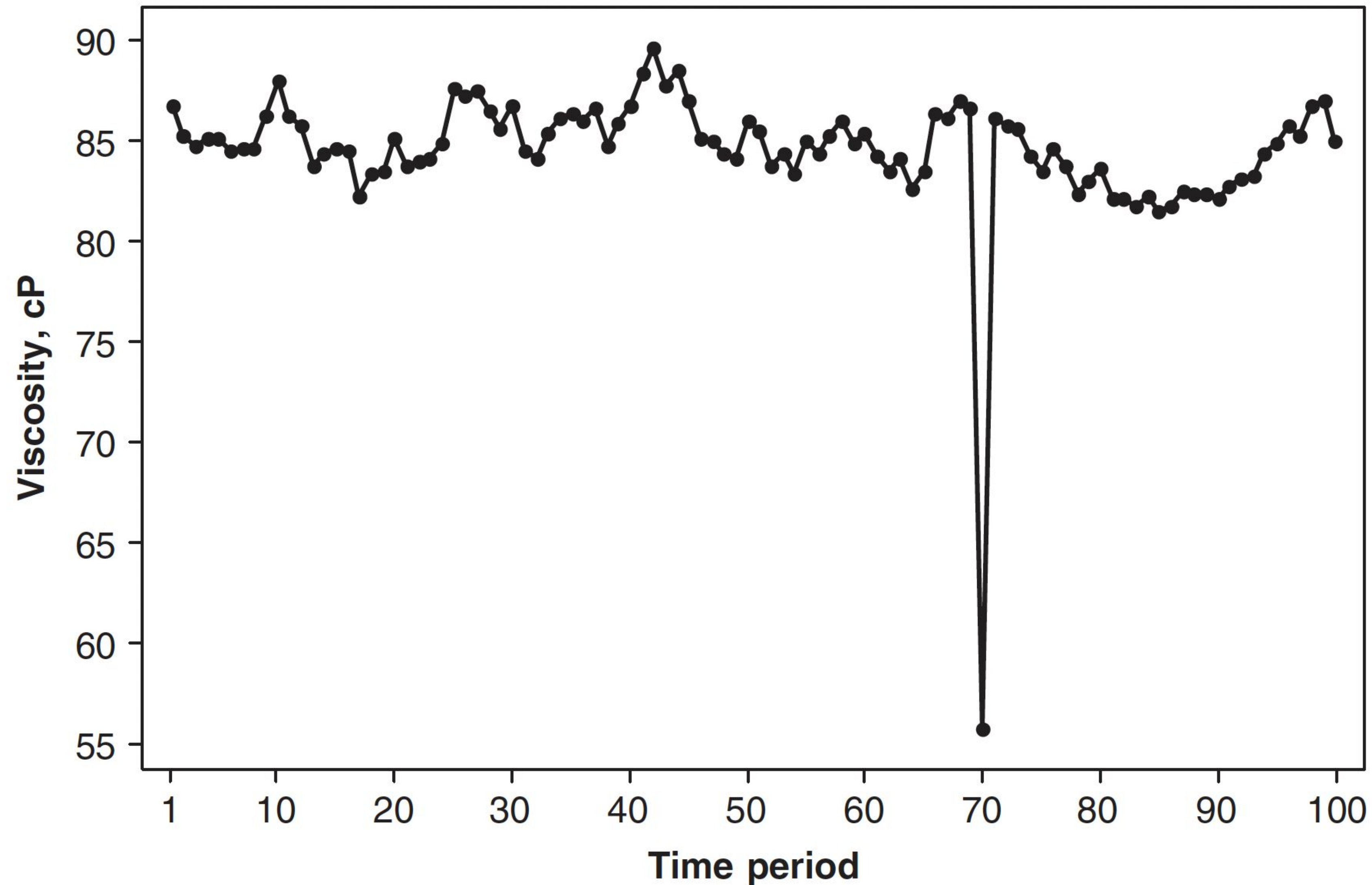


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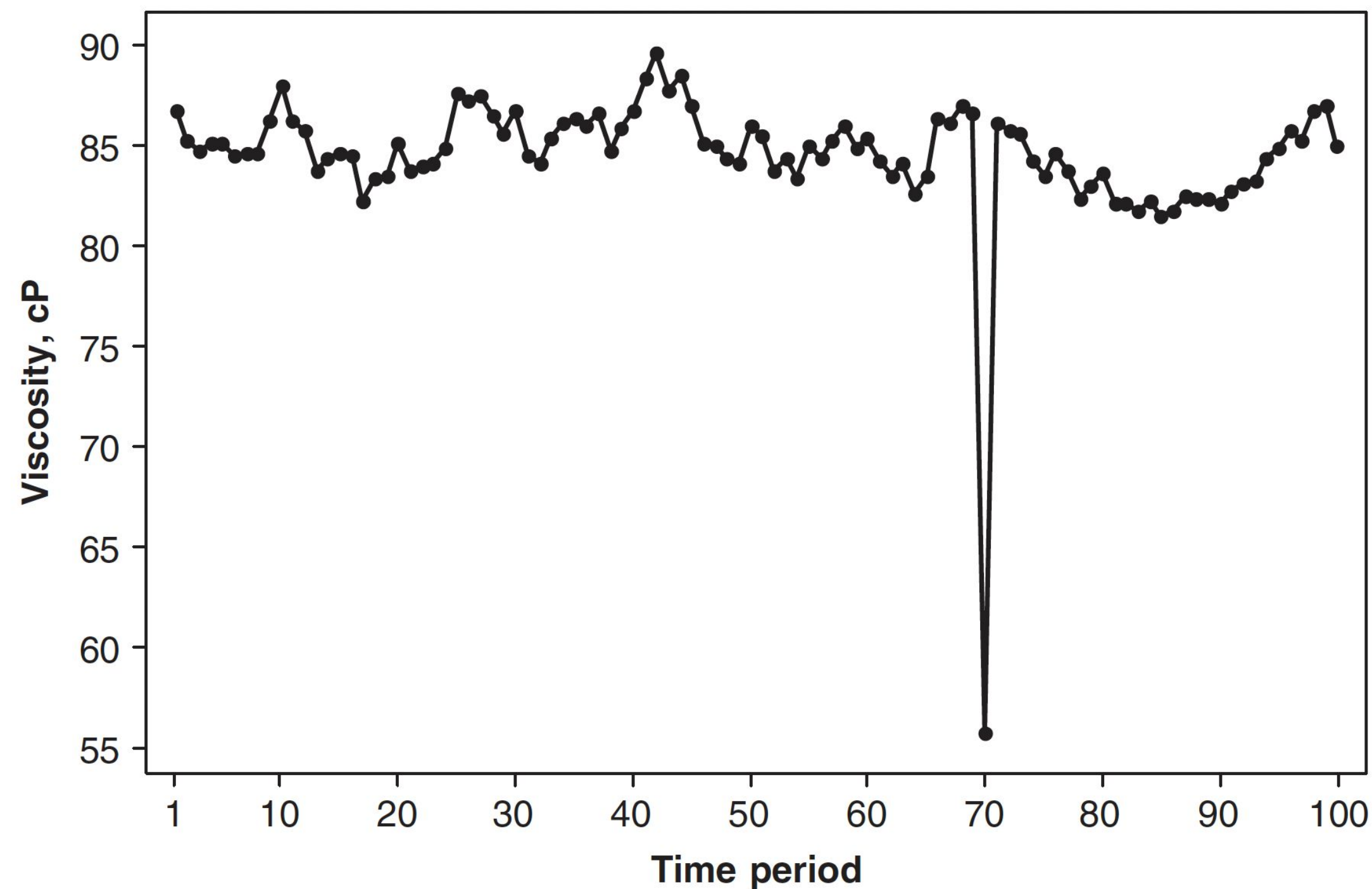


- An unusual significant drop is due to limited availability resulting from a fire at one of the four production facilities

Example 11: Chemical process viscosity readings (atypical events)



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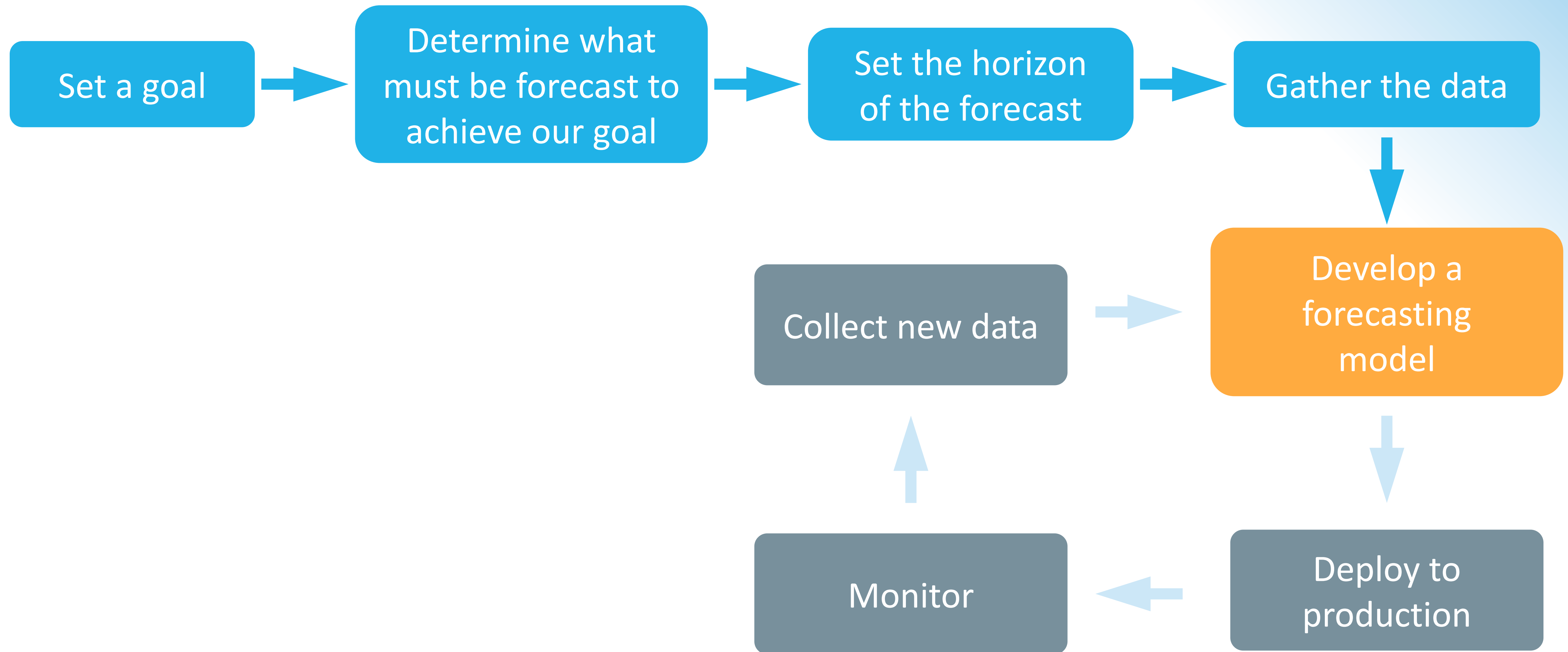


- Failure of the data measurement or collection system
- The cause was determined to be a malfunctioning sensor

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The Forecasting Process

The Forecasting Process



Thank you!